UADNet: A Joint Unmixing and Anomaly Detection Network Based on Deep Clustering for Hyperspectral Image

Wendi Liu, Yong Ma, Xiaozhu Wang, Jun Huang, Qihai Chen, Hao Li, and Xiaoguang Mei, *Member, IEEE*

***Abstract*— With the lack of sufficient prior information, unsupervised hyperspectral unmixing (HU) has been a preprocessing step in the hyperspectral image (HSI) processing pipeline, which can provide the types of material and corresponding abundance information of HSI, to further provide assistance for downstream higher level semantic tasks to overcome the limitation caused by mixed pixels. However, the unmixing results obtained by current unsupervised HU methods are unstable and unprecise under the guidance of the least reconstruction error (RE), which have no consistency with the performance of high-level tasks. To solve this problem, this article takes the hyperspectral anomaly detection (HAD) as an entry point and proposes a novel algorithm based on deep clustering which can jointly perform HU and HAD in an end-to-end manner. A mutual feedback mechanism is formed between the upstream HU process and the downstream HAD process, and through joint optimization, both two tasks can achieve relatively good performances. However, the low dimensional abundance has a limited representation, which may lead to the increase of false alarm rate. To overcome this limitation, the principal components (PCs) of HSI are fused with the abundance to enhance the representation ability. Moreover, we use the reweighted reconstruction loss strategy to enhance the role of anomalies in the HU process. Experiments performed on several real datasets verify the rationality and superiority of the proposed UADNet algorithm.**

***Index Terms*— Anomaly detection, deep clustering, high-level task, hyperspectral unmixing (HU), joint optimization.**

I. Introduction

HYPERSPECTRAL image (HSI) has the 3-D data structure, usually with dozens to hundreds of bands from visible light to infrared light, which provides more detailed spectral information than RGB image and multispectral image [1]. Due to the high resolution in spectral dimension, HSI has been widely applied in many remote sensing fields, such as classification [2], [3], [4], [5] and target detection [6], [7], [8]. However, limited by the low spatial resolution, mixed pixels commonly existed in the HSIs, which influences the performance of many high-level semantic tasks such as detection and classification. Hyperspectral unmixing (HU) is an important preprocessing step to solve the aforementioned problems, which tries to decompose the measured pixels into several pure materials (endmembers) and abundances relative to them [9].

Depending on whether the endmember information is known, HU methods can be divided into supervised HU and unsupervised HU. Affected by various environmental factors such as atmospheric effects and lighting conditions, it is difficult for the supervised algorithm to directly obtain the actual endmember spectrum that matches the real scene. Its complete and standardized spectra library is difficult to establish effectively. Hence, more works concentrate on the unsupervised HU algorithm. However, most unsupervised HU methods try to find the solution under the guidance of the least reconstruction error (RE) criterion with some prior knowledge, which is not directly related to the real distribution of various materials. Hence, without the reference to ground truth, the unmixing result obtained in this way is unstable and imprecise, and it is difficult to evaluate its quality.

In the HSI processing pipeline, HU can be regarded as the preliminary image understanding process, which can provide the types of material and corresponding abundance information of HSI, to further provide assistance for downstream higher level semantic tasks to overcome the limitation caused by mixed pixels. For example, in hyperspectral anomaly detection (HAD) fields, HU can produce differential background and anomaly abundance, providing discriminative criteria for the detection process. For subpixel mapping (SPM) tasks, the results are even more strongly dependent on the quality of the abundance map. These have been proven to be effective in some works [10], [11], [12]. However, the improvements of higher level tasks should rely on stable and precise unmixing results, which are difficult to solve with current unsupervised unmixing algorithms. That's because, in the design of the current HU methods, the relevance to these high-level tasks is rarely considered. Without other guidance, unmixing results

Manuscript received 11 September 2023; revised 13 January 2024; accepted 12 February 2024. Date of publication 14 March 2024; date of current version 21 March 2024. This work was supported by the Natural Science Foundation of Guangdong Province, China, under Grant 2023A1515012834. *(Corresponding authors: Xiaozhu Wang; Xiaoguang Mei.)*

Wendi Liu, Yong Ma, Jun Huang, and Xiaoguang Mei are with the Electronic Information School, Wuhan University, Wuhan 430072, China (e-mail: lwd2018_360@whu.edu.cn; mayong@whu.edu.cn; junhwong@whu.edu.cn; meixiaoguang@gmail.com).

Xiaozhu Wang is with the School of Data Science, The Chinese University of Hong Kong at Shenzhen, Shenzhen 518172, China (e-mail: wangc0104@163.com).

Qihai Chen is with the Tsinghua Innovation Center in Zhuhai, Zhuhai, Guangdong 519000, China (e-mail: chuangqi_cqh@163.com).

Hao Li is with the College of Mathematics and Computer Science, Wuhan Polytechnic University, Wuhan 430023, China (e-mail: lihao@whpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3375934

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

obtained by the least RE criterion are not consistent with the performance of high-level tasks. For example, for the HAD task, the core problem is to distinguish anomalies from the background pixels. Feedback to the HU task, how to make the background and anomaly more distinguishable in endmembers or abundance features is the key to better helping the HAD task, and the least RE is not necessarily the most significant criterion for HU. In this case, the performance of the high-level task can be a good metric for unmixing evaluation. Therefore, we hope to find a framework that can guide the HU and high-level tasks mutually, use the high-level task as a criterion to constrain the HU, and finally achieve the purpose of improving the performances of both.

Currently, only a few works have investigated the relevance between HU and high-level semantic tasks. For HAD, Qu et al. [10] performed this task in the abundance domain. The abundances were first obtained by the HU method MVCNMF [13]. Then, a dictionary based on the abundances was constructed to better represent the highly correlated background and the sparse anomaly. Finally, they performed the low-rank matrix decomposition to separate the background and anomaly. Song et al. [11] also adopted the abundance representation to perform HAD. Different from Qu's method, the abundance was extracted by the convolutional neural network. Both of these two algorithms use the results of HU to help HAD, but in the specific implementation, HU and HAD are conducted separately and sequentially according to their respective objectives rather than mutual constraints, which makes it easy to form cumulative errors. For SPM, Song et al. [12] proposed a joint HU and SPM framework based on multiobjective optimization. Such a joint framework can reduce the unmixing error propagated to SPM. The multiobjective optimization can ensure both the HU task and SPM task reach a relatively good result.

As the advantages of deep learning in the field of hyperspectral image processing become more and more prominent, there is still a lack of end-to-end joint HU and high-level semantic task framework. Especially for HAD, a typical high-level semantic task closely related to HU, to the best of our knowledge, no deep learning models have been proposed or studied yet for this type of multitask processing. The most related to our work is [14], which proposes an end-to-end framework for HAD. This work jointly optimizes the detection process and feature learning to improve the performance of HAD. However, the process of feature learning has no physical meanings, so [14] is still a separate HAD work.

In this article, we take an interest in the interplay between HU and HAD and investigate a new perspective for jointly optimizing the unmixing and anomaly problems in an end-to-end deep learning framework. It faces two main questions as follows.

1) The current HU and HAD are conducted separately, which makes it easy to produce cumulative errors. The HU operation does not improve the detector and sometimes even degrades it. How to design an effective mutual feedback mechanism to eliminate this cumulative error is a problem worth considering.
2) Unmixing can be regarded as a special feature learning process. Can HU be combined with the existing end-to-end HAD framework [14] to solve the problem of insufficient physical meaning? And whether this combination is conducive to the optimization of the two tasks?

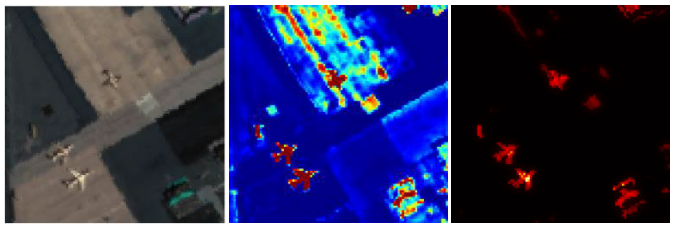
Motivated by these questions, this article proposes an unsupervised end-to-end joint HU and HAD network, named UADNet. Inspired by the basic idea of a deep clustering network, the HU process can be regarded as the process of feature learning, that is, learning the mapping from the original high-dimensional HSI to the low-dimensional abundance, while the HAD process can be regarded as the process of clustering, by clustering the low-dimensional features to identify anomalies. These two processes can be unified in the deep clustering network.

In our proposed UADNet, the HU process is performed by a convolutional autoencoder (AE), and the HAD process is performed by a clustering network. Different from the existing end-to-end clustering-based HAD algorithm [14], we impose unmixing constraints on the feature learning phase of the network, which makes the learned low-dimensional features have the physical meaning of abundance. As shown in Fig. 1, the abundance map has obvious similarity with the detection map. This is because the distribution of the anomalies can often be reflected by the distribution of the corresponding abundance. Hence, the physically-meaningful abundance features learned by unmixing constraints will undoubtedly help us better identify anomalous targets, which is also consistent with our intuition. Moreover, the HU process and HAD process are optimized jointly. A mutual feedback mechanism is formed between the two processes through the joint objective function. The accuracy of unmixing will directly affect the performance of clustering and anomaly detection. Meanwhile, the clustering loss of the HAD process can be regarded as a regularization term in the unmixing process, to guide similar pixels to learn similar structures in the abundance domain. Such a mutual feedback mechanism can eliminate the cumulative error from HU to HAD so that both two unsupervised tasks can achieve relatively good results. However, the low dimensional abundance has a limited representation, which may lead to the increase of false alarm rate. To overcome this limitation, the principal components (PCs) of HSI are fused with the abundance to enhance the representation ability. Then, the augmented features obtained by abundances and PCs are clustered by the clustering network. In addition, due to the imbalance in the number of background and anomaly samples, we use the reweighted RE as the loss of unmixing net to enhance the role of anomalies in the HU. In the detection phase, we use a Gaussian mixture model (GMM) to estimate the distribution of all pixels and perform the density estimation. The anomalous degree of each pixel can be calculated by the GMM probability density function (pdf).

In general, the major contribution of our algorithm can be summarized as follows.

1) We propose a novel algorithm UADNet which can jointly process HU and HAD two unsupervised tasks.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 1. (a) Pseudocolor image of SanDiego dataset. (b) Abundance of anomalies. (c) Detection result.

This algorithm unifies the upstream HU task with the downstream HAD task in an end-to-end manner. Moreover, we design the joint objective function so that the two tasks can be jointly optimized through the mutual feedback mechanism to achieve relatively good performances simultaneously.

2) In order to avoid the limited representation ability of low dimensional abundance, the PCs of HSI are fused with the abundance to enhance the representation ability. The PCs can be regarded as a supplement to the abundance feature which can increase more discriminative criteria to identify the anomalies from backgrounds.
3) Due to the imbalance in the number of background and anomaly samples, we use the reweighted RE as the loss of unmixing net to enhance the role of anomalies in the HU process. We increase the weight of the reconstruction loss of anomalous parts so that the anomalies can be reconstructed more accurately, which is beneficial to the subsequent HAD process.

The rest of this article is arranged as follows. Section II briefly introduces the HU method based on AE and the density estimation-based HAD method. In Section III, the proposed UADNet algorithm will be described in detail. Extensive experiments demonstrating the rationality and superiority of the proposed UADNet algorithm are conducted in Section IV. Section V draws the discussion and conclusion of our article.

II. Related Work

A. HU Based on LMM and AE

The linear mixture model (LMM) and nonlinear mixture model (NLMM) are the two main existing models used in HU [15]. LMM assumes the mixed pixel is the linear combination of endmembers while NLMM additionally considers the nonlinear interactions between pure spectrum [16]. For most scenarios, LMM can simulate the actual spectral mixing phenomenon simply and accurately. Hence, it's still the most widely studied and applied model in the HU community. The HU algorithm explored in this article is based on LMM.

According to LMM, the observed mixed pixels $\mathbf{Y} \in \mathbb{R}_+^{B\times P}$ are combined with endmembers $\mathbf{M} \in \mathbb{R}_+^{B\times R}$ and the corresponding abundances $\mathbf{A} \in \mathbb{R}_+^{R\times P}$. Then, LMM can be formulated as follows:

$$\mathbf{Y} = \mathbf{MA} + \mathbf{N}, \quad \mathbf{M}_{br}, \mathbf{A}_{rp} \geq 0, \quad \sum_{r=1}^{R} \mathbf{A}_{rp} = 1 \quad \forall p \tag{1}$$

where $\mathbf{N} \in \mathbb{R}_+^{B\times P}$ denotes the addictive model errors. $\{B, R, P\}$ denotes the number of bands, endmembers, and pixels, respectively. In order to satisfy the actual physical meaning, the endmember matrix is subjected to nonnegative constraint, and the abundance matrix is subjected to nonnegative constraint (ANC) and sum-to-one constraint (ASC) simultaneously.

Many classical HU algorithms have been proposed over the past few decades. They include geometry-based methods [17], [18], statistic-based methods [19], [20], sparse regression-based methods [21], [22], and nonnegative matrix decomposition (NMF)-based methods [23]. These algorithms rely on the design and extraction of handcraft features and are driven by a specific model, which can only adapt to a part of preassumed scenarios.

Recently, deep learning-based HU algorithms have attracted more attention for its powerful feature learning ability. The AE network, usually as the backbone network for HU, consists of two main parts: the encoder layer maps the HSI to a low-dimensional feature (abundance) while the decoder layer reconstructs the HSI with the corresponding basis (endmembers) [24]. For instance, Palsson et al. [25] used a deep AE (DAEU) to unmix the HSI and discussed HU performances influenced by different object functions. To incorporate spatial information in the HU process, a convolutional AE for HU (CNNAEU) [26] was proposed. The convolutional layer replaced the fully connected layer as the main structure of the encoder. Jin et al. [27] proposed an adversarial AE (AAE) for HU to solve the insufficient utilization of abundance prior. The AAE net matched the aggregated posterior of the abundance with a certain prior distribution to correct the weight of unmixing and get better performance. Yu et al. [28] proposed a multistage convolutional AE network (MSNet) to fuse the multiscale features so that the local details and global information are preserved simultaneously. The stability of HU was also improved meanwhile compared with conventional single-stage AE.

In AE architecture, the encoder $\mathcal{E}$ can encode the input data into a lower dimensional representation. The decoder $\mathcal{D}$ maps the lower dimensional representation to the reconstructed data by minimizing the REs. When the decoder is linearly activated, the weight of the decoder can be figured as endmember and the low-dimensional feature is the corresponding abundance, as shown in Fig. 2. Then, the mathematical form of HU can be expressed as

$$\mathbf{A} = \mathcal{E}(\mathbf{Y}; \theta_{\text{en}}) = \sigma(\mathbf{W}_{\text{en}}\mathbf{Y} + \mathbf{B}_{\text{en}}) \tag{2}$$

$$\hat{\mathbf{Y}} = \mathcal{D}(\mathbf{A}; \theta_{\text{de}}) = \mathbf{MA} \tag{3}$$

where $\mathbf{W}_{\text{en}}$ and $\mathbf{B}_{\text{en}}$ denote the weight and bias of encoder $\mathcal{E}$. σ denotes the active function of hidden layers.

B. Hyperspectral Anomaly Detection

In general, the HAD methods can be divided into two categories: 1) density estimation-based methods and 2) representation-based methods.

Density estimation-based methods assume that the background pixels $\mathbf{Y}_{\text{bkg}}$ are sampled from a certain distribution $f(\mathbf{Y}_{\text{bkg}}; \boldsymbol{\Theta})$, while the abnormal pixels deviate from the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

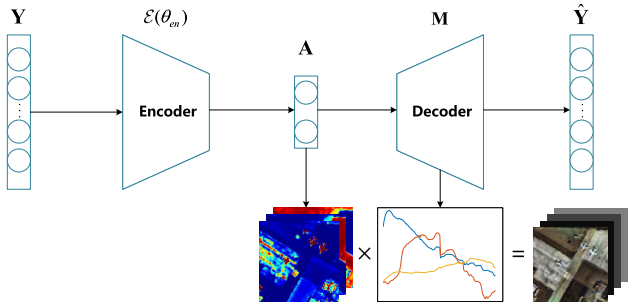


Fig. 2. AE-based unmixing network. **A** represents the abundance obtained from the encoder, and **M** represents the decoder weights, which can be seen as the extracted endmembers. Then, the reconstruction of HSI can be formulated as (3).

distribution. The parameters $\boldsymbol{\Theta}$ can be estimated via maximum-likelihood estimation, which can be formulated as follows:

$$\boldsymbol{\Theta} = \arg\max_{\boldsymbol{\Theta}} \sum_{p=1}^{P} \log f(\mathbf{y}_p; \boldsymbol{\Theta}). \tag{4}$$

Then, we can determine whether a pixel is normal or abnormal by measuring the degree of deviation. The famous HAD method, Reed–Xiaoli (RX) algorithm [6], is based on the assumption that the background pixels obey a Gaussian distribution. The degree of deviation is measured by calculating the Mahalanobis distance from every pixel to the mean value. Then, several variants of the RX algorithm have been proposed, such as local RX (LRX) [29], subspace RX (SSRX) [30], and kernel RX (KRX) [8].

However, the distribution of background may be more complicated in real scenarios. Thus, many works try to search for a more robust representation of the background. For example, CKRX [7] group pixels into many clusters so that there are multiple centers of backgrounds. Also, the clustering strategy decreases the computational burden of eigendecomposition, making CKRX perform more efficiently than KRX while the detection performance does not change so much.

Representation-based algorithms reconstruct the background from a global perspective under the assumption that anomalies usually behave differently from the background, usually assuming the background has the low-rank property and the anomaly has the sparsity property, then the background and anomaly are separated by low rank and sparse matrix decomposition (LRaSMD). Such typical works are LRASR [31] and LSMAD [32]. Li et al. [33] proposed the LTDD algorithm, taking tensor decomposition directly to maintain the 3-D structure of HSIs. Some recent works also try to construct the background dictionary to enhance the diversity of background [34], [35], [36].

Due to the good representation ability of the GMM, it is widely used in anomaly detection algorithms [14], [37], [38], [39]. In this article, we use GMM to implement density estimation. GMM can divide data points into multiple clusters and use a weight to represent the probability that a point belongs to a certain cluster. The distribution of all points is expressed as a weighted sum of multiple Gaussian distributions. The pdf of GMM is as follows:

$$f(\mathbf{y}; \boldsymbol{\Theta}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{y}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$
$$\boldsymbol{\Theta} = \{\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}\} \tag{5}$$

where π_k, $\boldsymbol{\mu}_k$, and $\boldsymbol{\Sigma}_k$ denote the weight, mean, and covariance of the kth cluster, respectively. After clustering the input data points, the anomalies will become outliers away from these clusters.

III. Proposed Methodology

The architecture of our proposed algorithm is shown in Fig. 3. The unmixing network is a convolutional AE. We input the HSI $\mathbf{Y}$ into the unmixing net and obtain the low-dimensional abundance representation $\mathbf{A}$ through the encoder. The abundance is then linearly decoded to obtain the reconstructed HSI $\hat{\mathbf{Y}}$, and the weight of the decoder is obtained as the endmember $\mathbf{M}$. In order to make the abundance map reflect the sparse characteristics of the abnormal target, we use the reweighted RE as the loss of the unmixing net. Due to the low dimension of the abundance feature and its limited representation ability, it may fail to distinguish the outliers by clustering abundance directly. Hence, we use PC analysis (PCA) to obtain the PCs of the original HSI and fuse them with the abundance features. Then, given the number of clusters K, we input the fused augmented features into the clustering net and obtain the mixture-component membership $\mathbf{c}_p \in \mathbb{R}^{K\times 1}$ of augmented vector $\mathbf{z}_p$. The density estimation is performed by the pdf of GMM. After calculating the pdf value of each pixel, the initial energy map is obtained. We use gray-scale transformation on the large-area connected regions in the energy map to suppress the background and obtain the final detection result. In the following parts, we specifically describe the framework of the unmixing net and clustering net in detail and then introduce the implementation details of density estimation. Finally, the details of the postprocessing are described.

A. Convolutional AE Framework for Unmixing

The convolutional AE is the backbone of the unmixing network as shown in Fig. 4. In general, AEs are simply composed of two parts: an encoder and a decoder.

1) Encoder: Blocks 1–4 form the encoder part $\mathcal{E}$ of the network for feature extraction and dimensional reduction of the input batch. Blocks 1–3 contain a series of operations, including the convolutional layer, activation function, batch normalization, and dropout. Batch normalization is applied to avoid the exploding or vanishing of gradients and accelerate the learning speed of the neural network. Considering the standard rectified linear unit (ReLU) function has a nonzero gradient for all inputs, we choose the leaky ReLU (LReLU) function as our activation function. The dropout layer can effectively alleviate the occurrence of overfitting and improve the generalization of the model.

To ensure the abundance meets the ANC and ASC conditions, a softmax operation is used after the convolution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

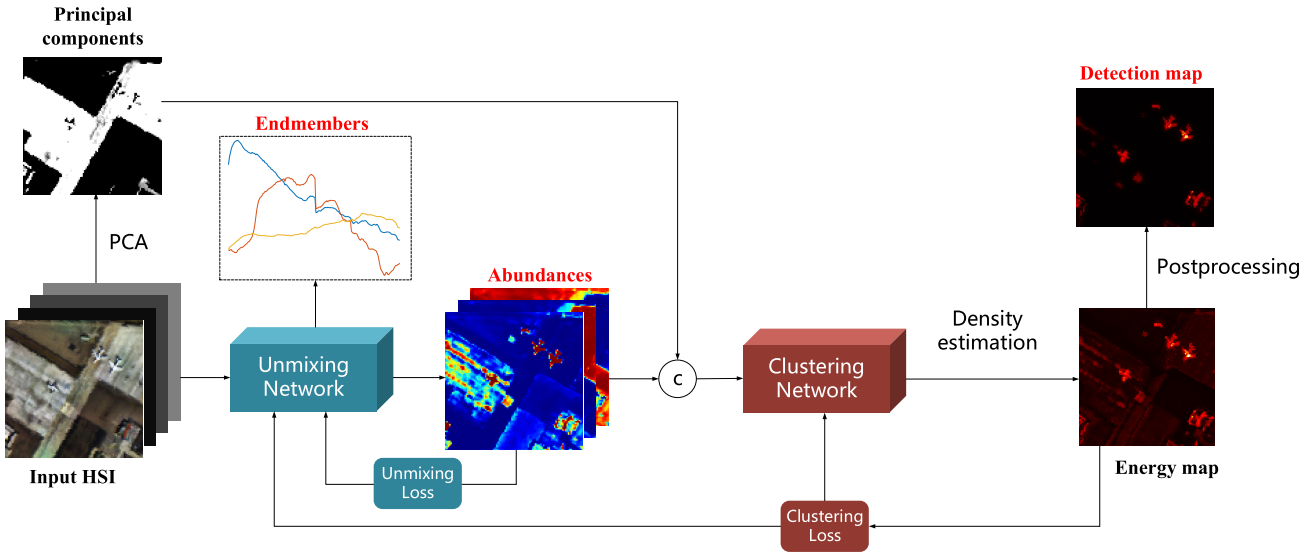


Fig. 3. Architecture of proposed algorithm. The proposed UADNet is combined with the unmixing network and clustering network. HU and HAD guide each other through the loss function.

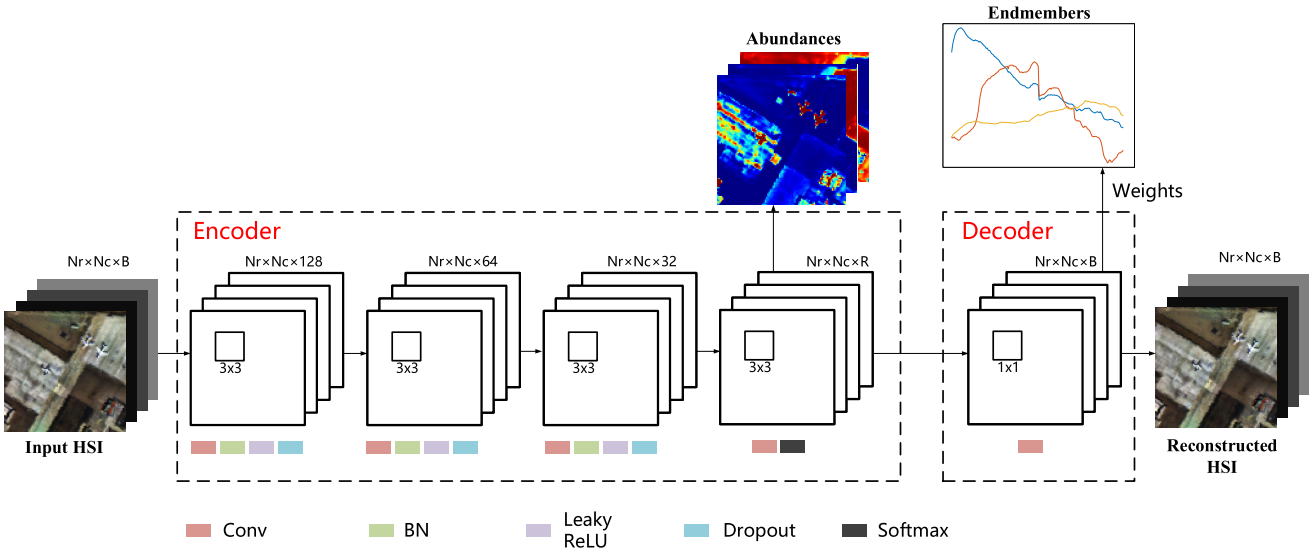


Fig. 4. Architecture of unmixing network. The abundances are the outputs of the last block of the encoder, and the endmembers are the weights of the decoder.

of the previous input. Then, given the input $\mathbf{Y}$, the output of the encoder $\mathcal{E}$ is as follows:

$$\mathbf{A} = \mathcal{E}(\mathbf{Y}; \theta_{\text{en}}) = \text{softmax}(\sigma(\mathbf{W}_{\text{en}}\mathbf{Y} + \mathbf{B}_{\text{en}})). \tag{6}$$

2) Decoder: Block 5 is the decoder $\mathcal{D}$ for reconstructing the learned abundance feature. According to the LMM assumption, block 5 uses a convolutional layer without bias to convolve the input and perform linear activation. The convolution kernel selects the size of 1×1, and the layer is equivalent to the fully-connected layer. Then, the weights of block 5 are regarded as the endmembers. The calculation process of the decoder $\mathcal{D}$ is as follows:

$$\hat{\mathbf{Y}} = \mathcal{D}(\mathbf{A}; \theta_{\text{de}}) = \mathbf{MA}. \tag{7}$$

B. Clustering Framework for Density Estimation

The architecture of the clustering network is shown in Fig. 5. The network has three main functions: feature fusion, clustering, and density estimation.

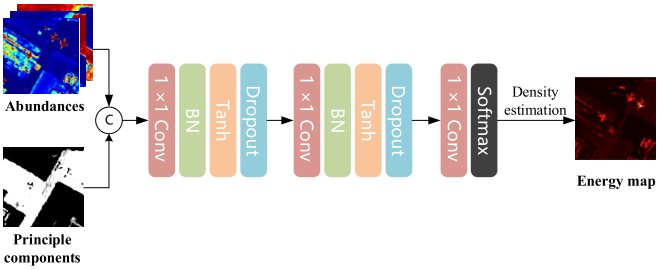
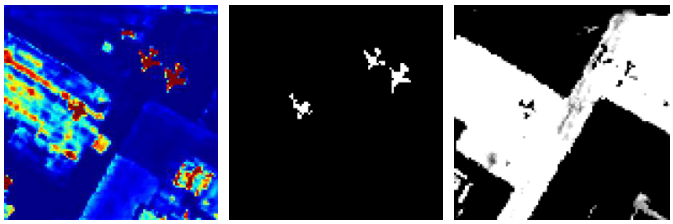


Fig. 5. Architecture of clustering network. The PCs, as the supplementary information, are concatenated with the abundances.

1) Feature Fusion: In general, the output of the hidden layer has a higher dimension to obtain a discriminative representation in the AE-based HAD algorithm. In this article, when the hidden layer outputs abundance, although it has a clearer physical meaning, it is difficult to directly form a criterion for anomaly detection due to its low dimension and insufficient representation ability. As shown in Fig. 6, taking

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 6. (a) Abundance map of SanDiego dataset. (b) Ground truth. (c) First PC of the SanDiego dataset.

SanDiego airport dataset as an example, Fig. 6(a) shows the abundance of airplanes, compared with ground truth Fig. 6(b), although the contour features of abnormal targets (airplanes) can be observed from the abundance map, there are also nonabnormal pixels with the same abundance pattern. If we only discriminate anomaly based on abundance, it will easily lead to a high false alarm rate of detection results, especially when the unmixing result is not ideal. Fig. 6(c) shows the first PC of HSI obtained by PCA, the edge features of airplanes can still be observed, while the nonabnormal pixels with the same abundance pattern in Fig. 6(a) are less appeared in Fig. 6(c). Hence, we integrate the PC features of HSI into the abundance features as a supplement. With the abundance $\mathbf{a}_p \in \mathbb{R}_+^{R\times 1}$ and PCs $\mathbf{y}_p^{\text{pca}} \in \mathbb{R}^{n_d\times 1}$, the augmented vector $\mathbf{z}_p$ of original spectra vector $\mathbf{y}_p$ is as follows:

$$\mathbf{z}_p = \begin{pmatrix} \mathbf{a}_p \\ \mathbf{y}_p^{\text{pca}} \end{pmatrix} \in \mathbb{R}^{(R+n_d)\times 1} \tag{8}$$

where n_d denotes the dimension of the spectra vector after PCA. In the implementation, we set n_d as 2. Finally, the augmented representation $\mathbf{z}_p$ is sent to the subsequent clustering network.

2) Clustering Process: The structure of the clustering net $\mathcal{C}$ is similar to the encoder part of the unmixing net mentioned in Section III-A. Here, we reduce one block and use all 1×1 convolution kernels. The activation function in blocks 1 and 2 is replaced by a hyperbolic tangent (Tanh). Then, output a membership vector $\mathbf{c}_p$ through the softmax operation in block 3 given the cluster number K

$$\mathbf{c}_p = \mathcal{C}(\mathbf{z}_p; \theta_{\text{cl}}) = \text{softmax}(\tanh(\mathbf{W}_{\text{cl}}\mathbf{z}_p + \mathbf{b}_{\text{cl}})) \tag{9}$$

where θ_{cl} refers to parameters of $\mathcal{C}$. $\mathbf{W}_{\text{cl}}$ and $\mathbf{b}_{\text{cl}}$ refer to the weights and bias of $\mathcal{C}$, respectively.

3) Density Estimation: After calculating the cluster membership of each vector through the network $\mathcal{C}$, given a batch of l augmented vectors $\{\mathbf{z}_1, \ldots, \mathbf{z}_l\}$, the parameters of GMM can be estimated as

$$\hat{\pi}_k = \sum_{p=1}^{l} \frac{c_{pk}}{l} \tag{10}$$

$$\hat{\boldsymbol{\mu}}_k = \frac{\sum_{p=1}^{l} c_{pk}\mathbf{z}_p}{\sum_{p=1}^{l} c_{pk}} \tag{11}$$

$$\hat{\boldsymbol{\Sigma}}_k = \frac{\sum_{p=1}^{l} c_{pk}(\mathbf{z}_p - \hat{\boldsymbol{\mu}}_k)(\mathbf{z}_p - \hat{\boldsymbol{\mu}}_k)^T}{\sum_{p=1}^{l} c_{pk}}. \tag{12}$$

The probability density of the spectra vector $\mathbf{y}_p$ can be derived according to (10)–(12)

$$p(\mathbf{z}_p) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{z}_p|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k). \tag{13}$$

Furthermore, the corresponding negative log likelihood of (13) can be formulated as $E(\mathbf{z}_p) = -\log p(\mathbf{z}_p)$. Based on the assumption that anomalies are always far from the clusters, the pdf value of anomalies is usually lower than normal pixels. Hence, we can construct an energy map $\mathbf{E} = \{E(\mathbf{z}_1), \ldots, E(\mathbf{z}_P)\}$ of P pixels. The higher $E(\mathbf{z}_p)$ is, the higher the probability that $\mathbf{y}_p$ is an abnormal pixel, as shown in Fig. 5.

C. Objective Function

The loss function of the proposed network consists of two main parts: unmixing loss and clustering loss. The unmixing loss tries to minimize the RE to obtain a reliable and physically meaningful unmixing result. The clustering loss minimizes the negative log-likelihood of the average probability density and obtains a high-confidence clustering result. At the same time, there is also a mutual feedback mechanism between the unmixing process and the clustering process. The accuracy of unmixing will directly affect the performance of clustering and anomaly detection. In addition, the clustering loss can be regarded as a regularization term in the unmixing process, to guide similar pixels to learn similar structures in the abundance domain.

1) Unmixing Loss: In this article, we take spectral angle distance (SAD) and mean square error (mse) as the criterion of RE, and use hyperparameter λ_{mse} to balance the contribution of these two items. The RE of $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$ is as follows:

$$\mathbf{RE}(\mathbf{y}_p, \hat{\mathbf{y}}_p) = \arccos\left(\frac{\langle \mathbf{y}_p, \hat{\mathbf{y}}_p \rangle}{\|\mathbf{y}_p\| \cdot \|\hat{\mathbf{y}}_p\|}\right) + \lambda_{\text{mse}}\|\mathbf{y}_p - \hat{\mathbf{y}}_p\|_2^2. \tag{14}$$

Assuming that there are P_{np} normal pixels $\mathbf{Y}_{\text{np}}$ and P_{ap} anomalous pixels $\mathbf{Y}_{\text{ap}}$ ($P = P_{\text{np}} + P_{\text{ap}}$) in the HSI $\mathbf{Y}$, in general, the average RE between $\mathbf{Y}$ and $\hat{\mathbf{Y}}$ is

$$\begin{aligned} \mathbf{RE}(\mathbf{Y}, \hat{\mathbf{Y}}) &= \frac{1}{P}\sum_{p=1}^{P} \mathbf{RE}(\mathbf{y}_p, \hat{\mathbf{y}}_p) \\ &= \frac{P_{\text{np}}}{P}\mathbf{RE}(\mathbf{Y}_{\text{np}}, \hat{\mathbf{Y}}_{\text{np}}) + \frac{P_{\text{ap}}}{P}\mathbf{RE}(\mathbf{Y}_{\text{ap}}, \hat{\mathbf{Y}}_{\text{ap}}). \end{aligned} \tag{15}$$

Since P_{ap} is usually much smaller than P_{np}, the RE of anomalous pixels contributes little to the overall RE, which will affect the unmixing accuracy of the abnormal part. Therefore, we readjust the weight of the abnormal part RE to balance the contribution of the abnormal part RE to the whole

$$\mathbf{RE}(\mathbf{Y}, \hat{\mathbf{Y}}) = (1 - \lambda_{\text{RE}})\mathbf{RE}(\mathbf{Y}_{\text{np}}, \hat{\mathbf{Y}}_{\text{np}}) + \lambda_{\text{RE}}\mathbf{RE}(\mathbf{Y}_{\text{ap}}, \hat{\mathbf{Y}}_{\text{ap}}). \tag{16}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

Theoretically, the hyperparameter λ_{RE} needs to be slightly larger than (P_{ap}/P). Hence, the reconstruction loss of unmixing can be formulated as

$$\text{Loss}_{\text{RE}} = \mathbf{RE}(\mathbf{Y}, \hat{\mathbf{Y}}). \tag{17}$$

In addition, we add a total variational loss of the endmember as a regularization term to obtain a smoother endmember spectral curve

$$\text{Loss}_{\text{MTV}} = \sum_{r=1}^{R} \sum_{b=1}^{B-1} \frac{1}{B-1} |m_{b+1,r} - m_{br}|. \tag{18}$$

Finally, the unmixing loss is as follows:

$$\text{Loss}_{\text{Unmixing}} = \text{Loss}_{\text{RE}} + \lambda_{\text{MTV}} \text{Loss}_{\text{MTV}} \tag{19}$$

where λ_{MTV} is the hyperparameter to balance Loss_{RE} and Loss_{MTV}.

2) Clustering Loss: According to (13), we use the negative log-likelihood pdf of the observed input samples in the augmented low-dimensional space to represent the loss of clustering. Furthermore, when the diagonal entries of covariance matrices degenerate to 0, trivial solutions are produced. Thus, by introducing a regularization term, small values on the diagonal entries can be penalized and trivial solutions can be avoided [14]. Then, the clustering loss can be formulated as

$$\begin{aligned} \text{Loss}_{\text{Clustering}} &= \lambda_E \frac{1}{P} \sum_{p=1}^{P} E(\mathbf{z}_p) + \lambda_C \sum_{k=1}^{K} \sum_{j}^{d} \frac{1}{\hat{\mathbf{\Sigma}}_{jj}} \\ &= \lambda_E \text{Loss}_{\text{Energy}} + \lambda_C \text{Loss}_{\text{Cov}} \end{aligned} \tag{20}$$

where d is the size of covariance matrix $\hat{\mathbf{\Sigma}}_{jj}$.

Combining (19) and (20), the total loss of our proposed network is as follows:

$$\text{Loss} = \text{Loss}_{\text{Unmixing}} + \text{Loss}_{\text{Clustering}}. \tag{21}$$

D. Postprocessing

After constructing the energy map $\mathbf{E}$ of anomaly detection through the network, we further refine it to obtain a more intuitive detection map. The postprocessing adopts the method in [14]. The main idea is to use the gray-scale transformation on the large area-connected region in the energy map. Due to the small size and low probability density of anomalies, the gray level is usually higher than that of the large area-connected background. The gray-scale transformation further widens the gap between anomaly and background to suppress the background and highlight targets.

Specifically, the energy map $\mathbf{E}$ is first subjected to Otsu binarization [40] to obtain the binary image $\mathbf{T}$. Then, we assign 0 to the connected component with an area larger than β in $\mathbf{T}$ and 1 to other regions to obtain the Boolean map $\mathbf{B}$ of $\mathbf{T}$. Therefore, we mark all the large-area connected regions. In the energy map $\mathbf{E}$, we perform a gray-scale transformation on these regions to obtain the final detection map $\mathbf{D}$.

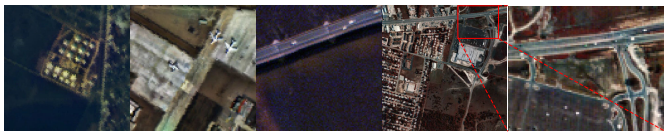


Fig. 7. Pseudocolor images of four datasets. (a) Coast dataset. (b) SanDiego dataset. (c) Pavia dataset. (d) Original HYDICE urban dataset. (e) Subimage of HYDICE dataset.

IV. Experiments

A. Dataset

Four real-world hyperspectral datasets, as shown in Fig. 7, are utilized to verify the accuracy and effectiveness of our proposed method.

1) Coast Dataset: The Coast dataset was acquired by the Airborne Visible/ Infrared Imaging Spectrometer (AVIRIS) sensor, which contains 100 × 100 pixels, each with 198 channels, covering a range of 450–1350 nm. Buildings of different scales are regarded as anomalies.

2) SanDiego Dataset: Acquired by the AVIRIS sensor, the SanDiego dataset has 189 spectral bands in total covering the wavelength ranging from 370 to 2510 nm and 100 × 100 spatial pixels. In this scene, hangars, parking aprons, and exposed soil constitute the main background, and three airplanes with 134 spatial pixels are regarded as anomalies.

3) Pavia Dataset: The Pavia dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor over Pavia city, which consists of 100 × 100 pixels and 102 spectral bands ranging from 430 to 860 nm. The vehicles are considered as anomalies containing 63 pixels.

4) HYDICE Dataset: Acquired by the hyperspectral digital imagery collection experiment (HYDICE) sensor over an urban area, CA, USA, the HYDICE dataset has 307 × 307 spatial pixels with 162 spectral channels. We cut a subimage of size 80 × 100 as our experimental dataset from the top-left point (0, 186), as shown in Fig. 7(e). The anomalies are vehicles of different sizes composed of 19 pixels in the urban scene.

B. Performance Evaluation Metrics

1) HU Metrics: Two commonly used evaluation metrics are introduced, including root-mean-square error (RMSE) and SAD, which are calculated by

$$\text{RMSE}_A = \frac{1}{P} \sum_{p=1}^{P} \sqrt{\frac{1}{R} \|\mathbf{a}_p - \hat{\mathbf{a}}_p\|^2} \tag{22}$$

$$\text{SAD}_M = \frac{1}{R} \sum_{r=1}^{R} \arccos\left(\frac{\langle \mathbf{m}_r, \hat{\mathbf{m}}_r \rangle}{\|\mathbf{m}_r\| \cdot \|\hat{\mathbf{m}}_r\|} \right). \tag{23}$$

If no ground-truth abundance or endmember is available, for example, for most real datasets used in HAD, we consider the pixel RE as the main criterion

$$\text{RMSE}_Y = \frac{1}{P} \sum_{p=1}^{P} \sqrt{\frac{1}{B} \|\mathbf{y}_p - \hat{\mathbf{y}}_p\|^2} \tag{24}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

$$\mathrm{SAD}_Y = \frac{1}{P}\sum_{p=1}^{P}\arccos\left(\frac{\langle \mathbf{y}_p, \hat{\mathbf{y}}_p\rangle}{\|\mathbf{y}_p\|\cdot\|\hat{\mathbf{y}}_p\|}\right). \tag{25}$$

2) HAD Metrics: The receiver operating characteristic (ROC) is widely used to evaluate the validity of detectors. Given the classification threshold τ, the true positive rate P_d and false positive rate P_f can be represented as

$$P_d = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{26}$$

$$P_f = \frac{\mathrm{FP}}{\mathrm{TN}+\mathrm{FP}} \tag{27}$$

where TP, TN, FP, and FN are denoted as the number of true positive, true negative, false positive, and false negative samples, respectively. We use ROC(P_d, P_f) curve and ROC(P_f, τ) curve to evaluate the detection effectiveness and the background suppression ability. The area under the curve (AUC) values of ROC(P_d, P_f) curve and ROC(P_f, τ) curve are calculated to quantify them.

Since the numbers of positive and negative samples are imbalanced in the HAD task, the precision–recall (PR) curve is also introduced to describe the performance of different detectors [41], [42]. Given the threshold τ, the precision and recall are calculated as

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{28}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{29}$$

Similarly, we use the AUC($P-R$) to quantify the performance of detectors. Larger AUC (P_d, P_f) and AUC ($P-R$) values indicate better detection performance, and a lower AUC (P_f, τ) value indicates better background suppression ability.

The Box-Whisker plot displays the variation in samples of a statistical population, which is nonparametric without any assumption of the underlying statistical distribution. The degree of dispersion and skewness in the data can be described by the spacing between the different parts of the box.

C. Sensitivity Analysis

In this section, we will analyze the sensitivity of the parameters on the performance of the proposed method. There are six significant parameters that control the entire HAD process. K is the number of Gaussian components for the clustering process. The hyperparameters $\{\lambda_{\mathrm{mse}}, \lambda_{\mathrm{RE}}, \lambda_{\mathrm{MTV}}\}$ in (14), (16), and (19) mainly affect the accuracy of HAD by controlling the HU process and $\{\lambda_E, \lambda_C\}$ in (20) control the clustering process to influence the HAD result. In this experiment, when the analysis of a certain parameter is performed, the other parameters are fixed as the corresponding optimal values as shown in Table I. The variation trend of AUC(P_d, P_f) scores for four datasets with each parameter is shown in Fig. 8.

1) Number of Gaussian Components K: The number of Gaussian components K can influence the result of clustering directly. When K is set to a very small value, the clustering net will not separate the anomaly pixels from the background pixels completely. In this experiment, K is ranging from 1 to 10. As shown in Fig. 8(a), when K is set to 5, the proposed detector can get good AUC scores on all datasets. But for the Pavia dataset, the scenario is relatively simple due to its well-regulated distribution of materials. A larger K will lead to error accumulation when calculating the pdf value. Hence, the optimal K for the Pavia dataset is smaller than other datasets. For the Coast dataset, we can observe when K is over 7, the AUC score degrades rapidly. Here, we speculate the reason is that the proportion of anomalies in this dataset is higher than in other datasets. With the increase of K, the background pixels are also divided into more clusters. It will cause the sparsity of anomalies to be diluted by these smaller clusters. Then, the background pixels in these smaller clusters are more likely to be recognized as anomalies, which leads to the degradation of the detector.

TABLE I
OPTIMAL PARAMETER SETTINGS FOR ALL DATASETS IN THE SENSITIVITY ANALYSIS

Parameter	K	λ_{MSE}	λ_{RE}	λ_{MTV}	λ_{E}	λ_{C}
Coast	5	0.5	0.2	0.1	0.01	0.0001
SanDiego	6	0.5	0.2	0.1	0.001	0.0001
Pavia	1	0.5	0.2	0.1	0.01	0.0001
HYDICE	4	0.5	0.1	0.1	0.01	0.0001

2) Hyperparameter $\lambda_{\mathrm{MSE}}, \lambda_{\mathrm{RE}}$, and λ_{MTV}: These three hyperparameters are closely related to the HU process, thus affecting the accuracy of HAD. The tradeoff hyperparameter λ_{mse} balances the contribution of the mse to the RE, λ_{RE} controls the contribution of the anomaly RE to the total RE, and λ_{MTV} balances the contribution endmember total variational regularization to the unmixing loss. The range of λ_{mse} and λ_{MTV} are set to $\{0, 10^{-4}, 5\times10^{-4}, 10^{-3}, 5\times10^{-3}, 10^{-2}, 5\times10^{-2}, 10^{-1}, 5\times10^{-1}, 1\}$, and λ_{RE} is varying from 0 to 1. As shown in Fig. 8(b)–(d), all datasets can get good performances when $\{\lambda_{\mathrm{mse}}, \lambda_{\mathrm{RE}}, \lambda_{\mathrm{MTV}}\}$ is set to $\{0.5, 0.2, 0.5\}$, and the AUC scores are relatively more robust to λ_{MTV}. The AUC score of the Pavia dataset is robust to changes in all three parameters, while λ_{RE} has a great impact on the Coast dataset and HYDICE dataset. When λ_{RE} is over 0.5, the AUC scores are even close to 0.5, which illustrates the detector almost loses the detection ability. We can speculate that for complex scenarios, the extraction of the background information has a great influence on HAD. When the λ_{RE} increases, the network pays more attention to the reconstruction of the anomalies and ignores the extraction of the background information, which leads to the destruction of the overall HU process of the whole HSI and makes the detector invalid.

3) Hyperparameter λ_E and λ_C: These two hyperparameters balance the contribution of clustering loss to the total loss. The ranges of λ_E and λ_C are set to $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ and $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, respectively. A larger λ_E and λ_C may affect the HU process proceeding toward a physically meaningful direction, generating meaningless unmixing results and making it difficult for the clustering network to extract useful information from the abundance map, while

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

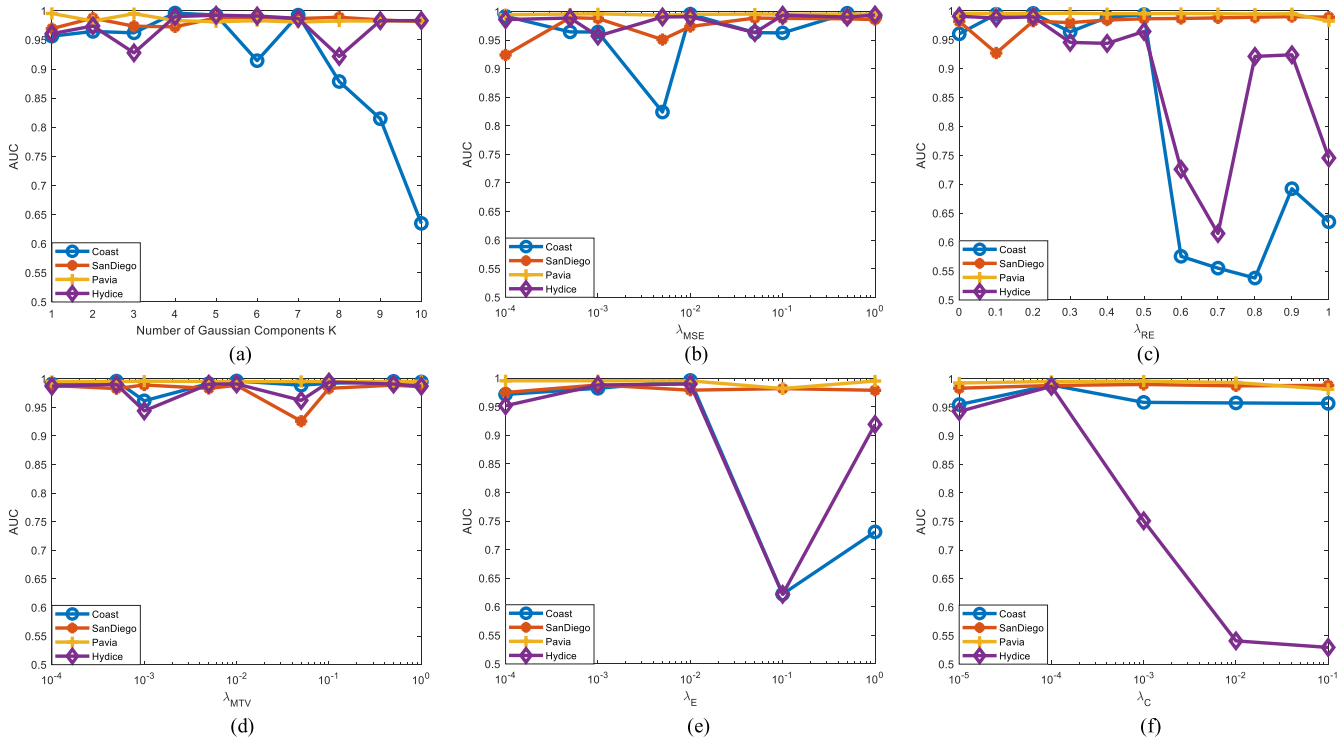


Fig. 8. Parameter sensitivity analysis of the influence on different datasets. (a) Number of Gaussian components K. (b) λ_{mse}. (c) λ_{RE}. (d) λ_{MTV}. (e) λ_E. (f) λ_C.

a smaller λ_E and λ_C cannot enable the clustering network to cluster the abundance features effectively. As illustrated in Fig. 8(e)–(f), the detection performance for Coast dataset and HYDICE degrades a lot while λ_E is over 0.01. λ_C also has a great impact on the HYDICE dataset, and the AUC score of the Coast dataset also decreases slightly when λ_C increases. As we mentioned before, the HU process will be affected while these two hyperparameters increase. The HAD accuracy relies more on the abundance of information for these two datasets. Hence, the detector dose not perform well when the HU network cannot output a relatively accurate result. The detector can perform well when the optimal settings of $\{\lambda_E, \lambda_C\}$ are selected as $\{10^{-2}, 10^{-4}\}$ according to Fig. 8(e)–(f).

D. Comparison Methods

In order to verify the effectiveness of the proposed algorithm, we selected some state-of-the-art methods and conducted comparative experiments of unmixing and anomaly detection. Before applying the experiments, we briefly introduce these methods as follows.

1) HU Algorithms: For HU comparison experiments, we chose seven algorithms as follows.

1) *MVCNMF [13]:* MVCNMF is a classical nonnegative-matrix-decomposition-based HU algorithm that conducts a minimum volume constraint to estimate the endmembers.
2) *SGSNMF [43]:* This method performs superpixel segmentation on HSI and mines the group sparsity properties in the superpixel.
3) *CNNAEU [26]:* CNNAEU is a typical AE-based HU algorithm that designs a convolutional AE to incorporate spatial information in HU.
4) *MiSiC-Net [44]:* This method incorporates a minimum simplex volume penalty term in the convolutional network to reduce the influence on endmember estimation when there are few pure pixels.
5) *CyCU-Net [45]:* CyCU-Net is capable of reducing the detailed and material-related information loss in the process of reconstruction by relaxing the original pixel-level reconstruction assumption to cycle consistency dominated by the cascaded AEs.
6) *PGMSU [46]:* PGMSU is a probabilistic generative model to address endmember variability. It allows fitting arbitrary endmember distributions through the nonlinear modeling capability of neural networks.
7) *MSNet [28]:* MSNet is a multistage convolutional AE network. Each subnetwork performs the HU process at different scales. The abundances of different scales are constrained by the edge consistency loss. The local details and global information are preserved simultaneously.

2) HAD Algorithms: We chose ten HAD algorithms as the comparison methods for detection experiments as follows.

1) *RX [6]:* The well-known RX algorithm assumes that the background obeys a Gaussian distribution and computes the anomaly score for each pixel by Mahalanobis distance.
2) *CKRX [7]:* This method is the generalization of KRX. The key idea is to group background pixels into clusters and just need to measure the distance between the testing pixel and the center of clusters.
3) *LSMAD [32]:* This method assumes that background has the low-rank property and anomaly has the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

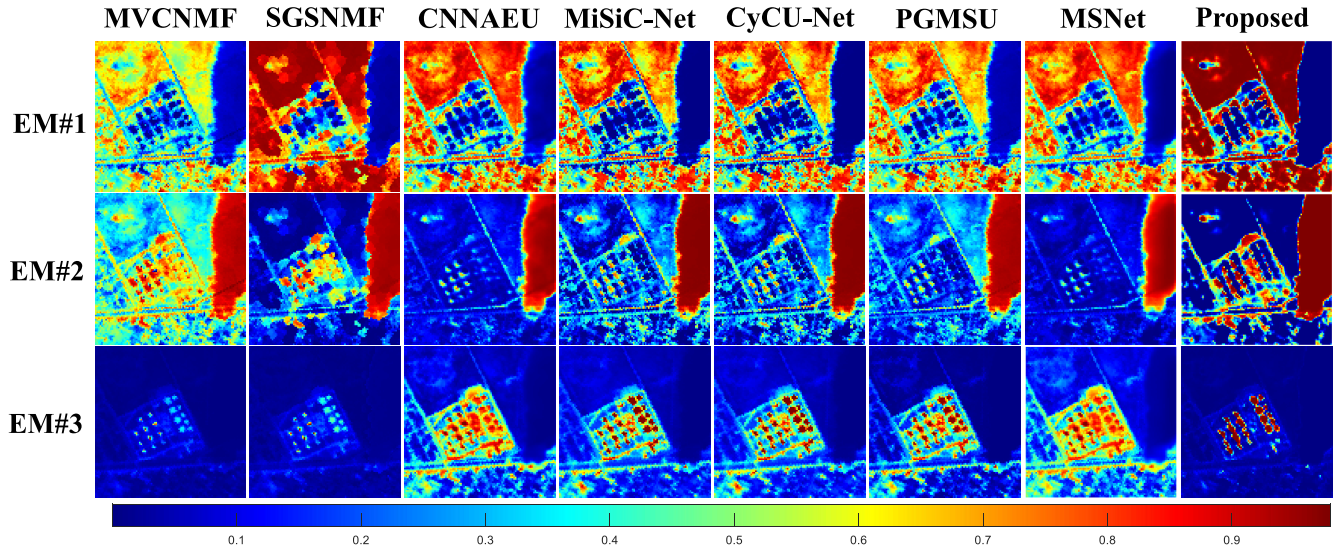


Fig. 9. Visual abundance maps of different methods on the Coast dataset.

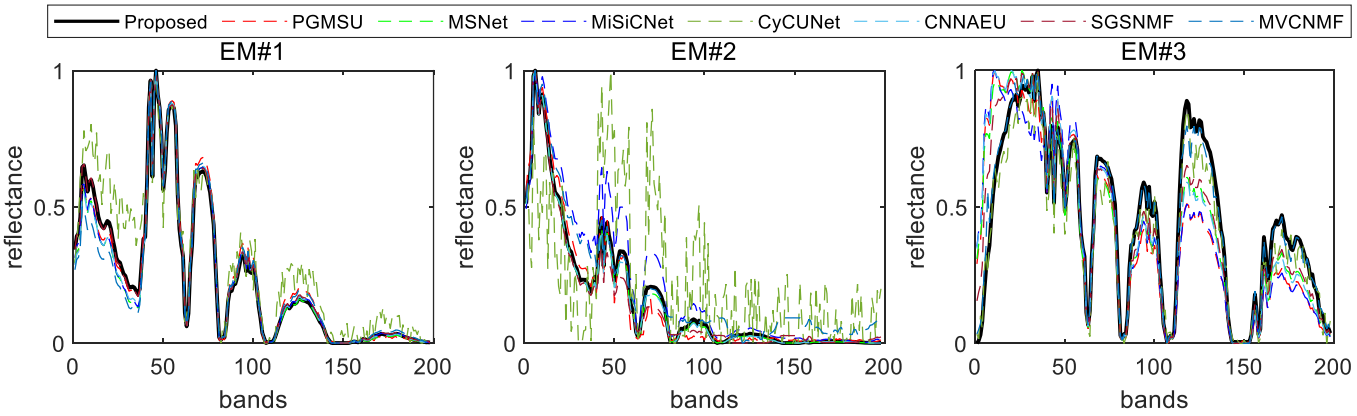


Fig. 10. Endmembers estimation of different methods on the Coast dataset.

sparsity property. The sparse matrix is separated by an optimization method.

4) *ADLR [10]:* This method considers that the abundance value of the anomaly cannot be represented. Unlike LSMAD, the algorithm performs sparse matrix decomposition in the abundance domain.
5) *LSMD-MoG [39]:* This method modifies the LSMD algorithm by modeling the sparse component as a mixture of Gaussian and using the variational Bayes algorithm applied to infer the model.
6) *RGAE [47]:* This method proposes a robust AE framework with $l_{2,1}$-norm which is robust to noise and anomalies during training and embeds a superpixel segmentation-based graph regularization term to preserve the geometric structure of HSI.
7) *SaFra [34]:* This method employs a framelet decomposition to characterize background representation coefficients and exploits saliency prior knowledge to measure each pixel's potential to be an anomaly.
8) *AHMID [35]:* This method designs different constraints for background and exception to improve their separability and constructs a mixed noise model to improve the anti-noise performance.
9) *DSR-ADNR [36]:* This method simultaneously performs noise removal and anomaly detection to control the influence of noise on the detection results, which is also based on the idea of the combination of high and low-level vision tasks.
10) *E2E-LIADE [14]:* This method designs an end-to-end model to simultaneously learn a low-dimensional representation of HSI and perform a density estimation to measure anomaly. The local structure information of HSI is fully utilized by a manifold regularization during HAD.

E. Unmixing Performance

For the proposed method, the hyperparameters $\{\lambda_{\text{mse}}, \lambda_{\text{RE}}, \lambda_{\text{MTV}}, \lambda_E, \lambda_C\}$ are set according to Table I. The hyperparameters of other comparison methods are set according to their original descriptions. The estimated abundance maps are demonstrated in Figs. 9, 11, 13, and 15. The estimated endmembers are demonstrated in Figs. 10, 12, 14, and 16. The quantitative results are displayed in Table II. **Bold** indicates the best result, underlined indicates the second best result.

Due to the small size and sparsity of anomalous targets, the abundance estimation of anomalies is more challenging

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

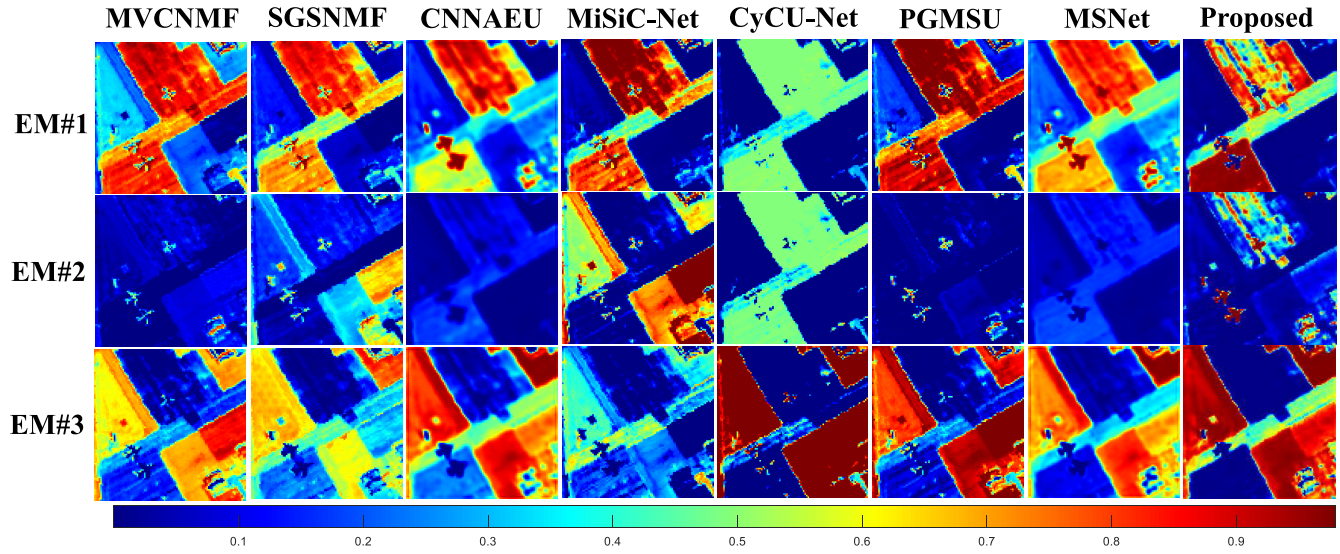


Fig. 11. Visual abundance maps of different methods on the SanDiego dataset.

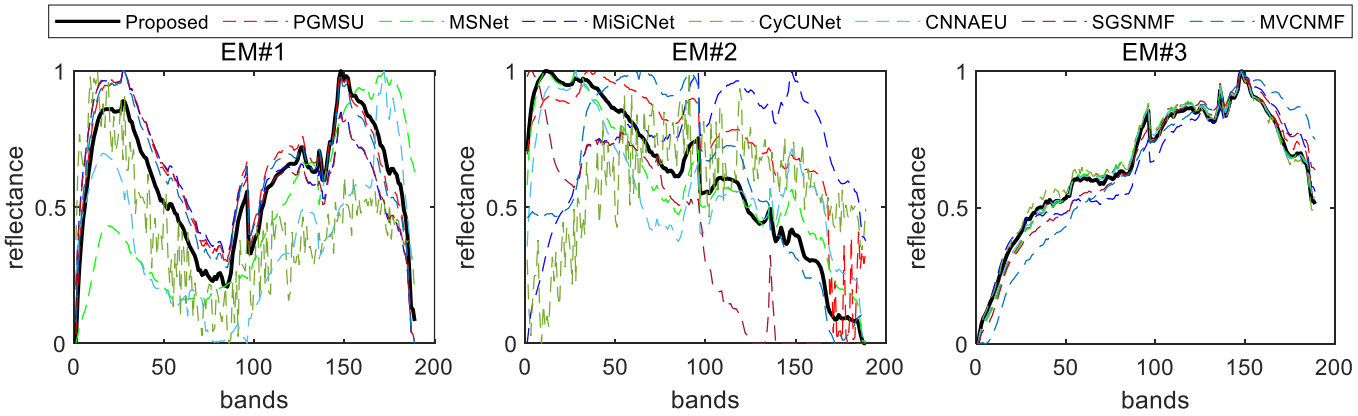


Fig. 12. Endmembers estimation of different methods on the SanDiego dataset.

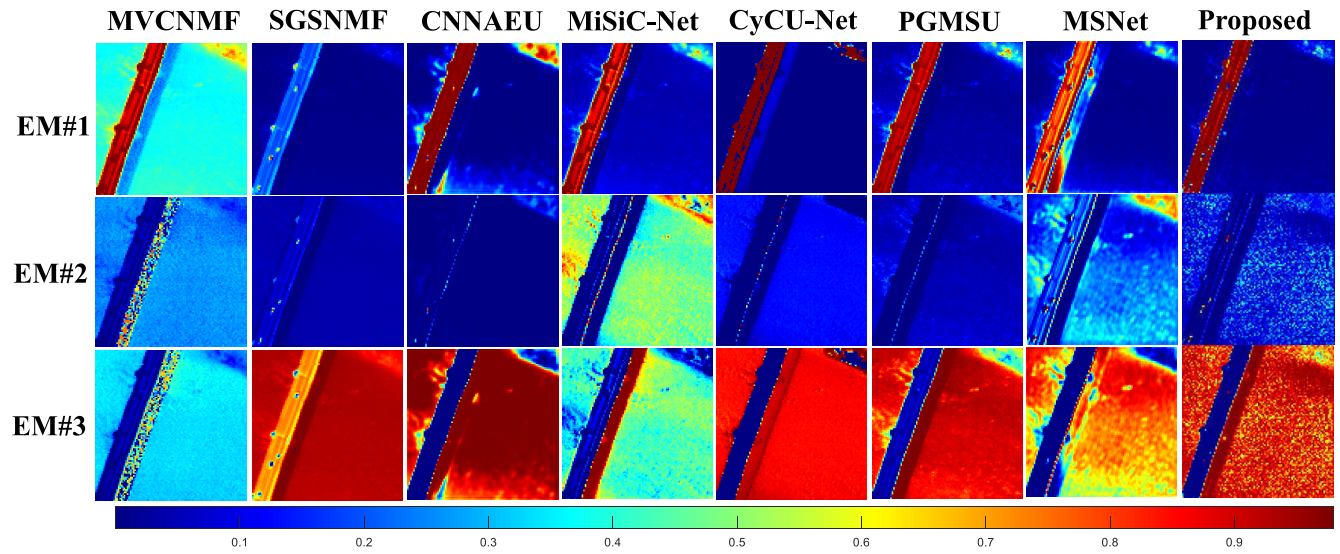


Fig. 13. Visual abundance maps of different methods on the Pavia dataset.

than other materials. There is no ground truth for endmember and abundance in the Coast, SanDiego, and Pavia datasets. So we only evaluate the REs according to (24) and (25) for these three datasets. However, a small value of the RE does not completely imply better unmixing performance. For unsupervised unmixing, the meaningful unmixing results should benefit high-level vision tasks. When the assumed high-level task is anomaly detection, we should pay more attention to the connection between the unmixing result and the anomaly detection task.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

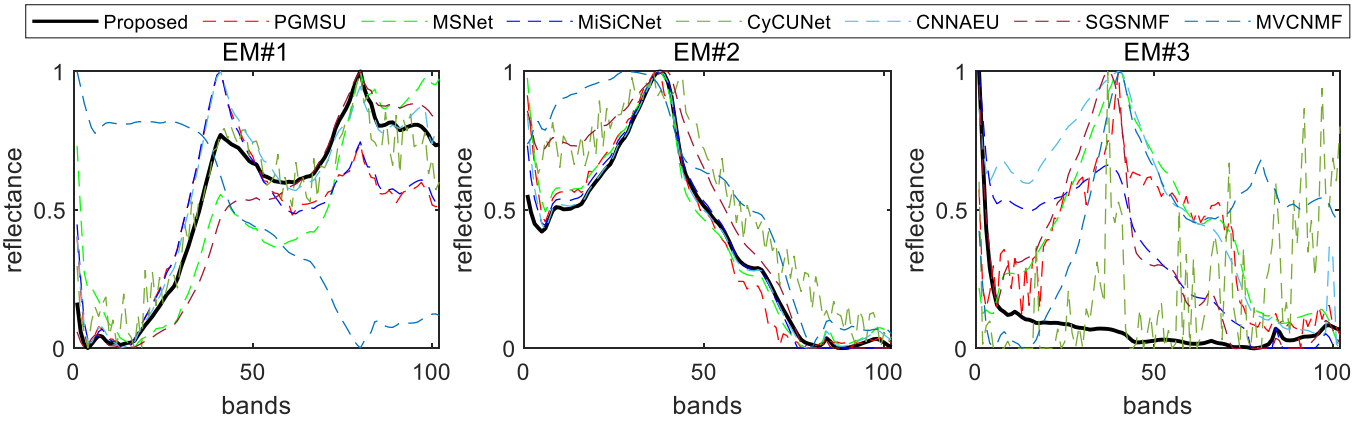


Fig. 14. Endmembers estimation of different methods on the Pavia dataset.

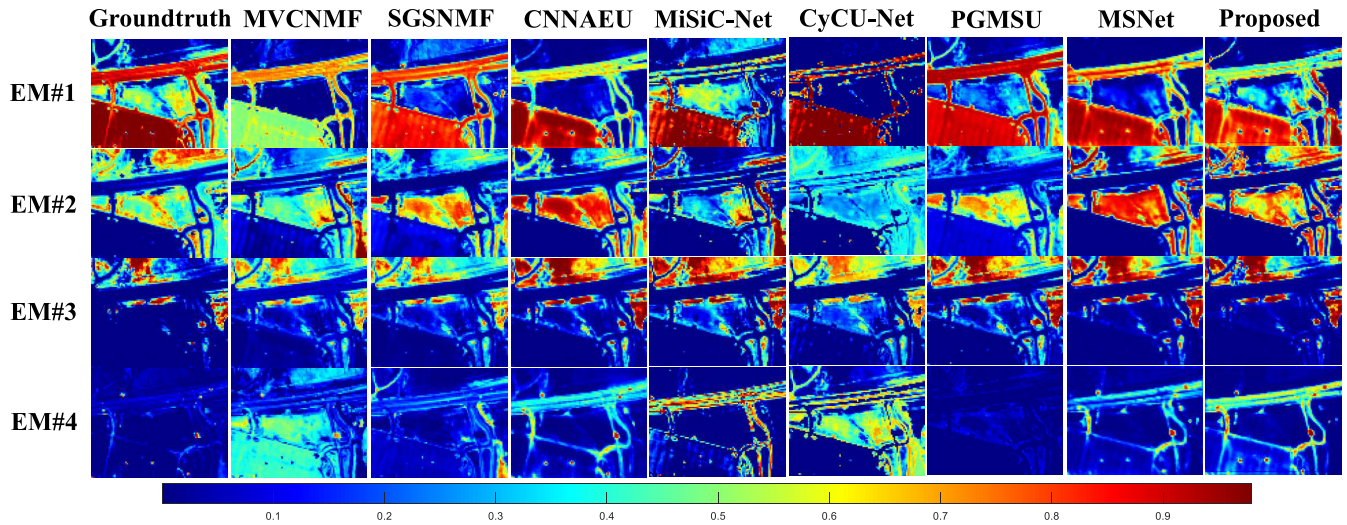


Fig. 15. Visual abundance maps of different methods on the HYDICE dataset.

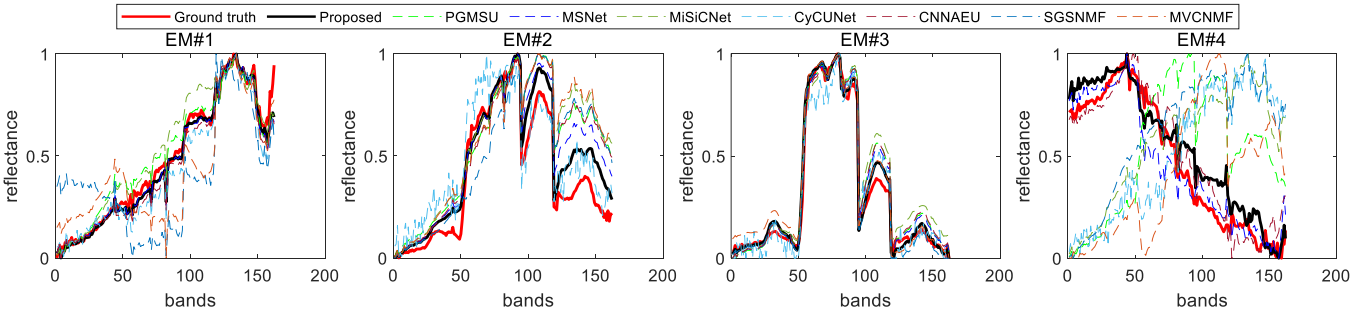


Fig. 16. Endmembers estimation of different methods on the HYDICE dataset.

The two traditional HU algorithms, MVCNMF and SGSNMF, have limited performances due to the lack of powerful feature extraction capabilities as neural networks. MVCNMF is also the preunmixing algorithm of HAD algorithm ADLR, the error caused by unmixing will directly affect the detection result, which will be discussed in the next part. The other deep learning-based algorithms, CNNAEU, MiSiC-Net, PGMSU, and MSNet, can get good reconstruction performances in the first three datasets. However, these RE-guided unmixing methods may cause overfitting, leading to uncertainty in unmixing results and loss of physical meaning. Especially for the abundance maps of anomalies, these methods do not fully catch the properties of these anomalous targets.

For our proposed method, the estimated abundance maps can significantly reflect the shape, edge, and other characteristics of the target, which can be observed in the abundance map of EM3, EM2, EM2, and EM4 in Figs. 9, 11, 13, and 15, respectively. In other words, we can even observe anomalies directly from the abundance maps. Moreover, the abundance values of anomalies estimated by our proposed method are nearer than 1 compared with other methods. It means our unmixing results are more beneficial to form a criterion for anomaly detection.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

TABLE II
UNMIXING PERFORMANCE FOR THE ALGORITHMS ON THE EXPERIMENTAL DATASETS

Dataset	Metrics	MVCNMF	SGSNMF	CNNAEU	MiSiC-Net	CyCU-Net	PGMSU	MSNet	Proposed
Coast	$\mathrm{RMSE_Y}$	0.0766	0.0558	**0.0253**	0.0400	0.0990	0.0381	0.0400	0.0595
	$\mathrm{SAD_Y}$	0.1752	0.1171	**0.0520**	0.0870	0.2370	0.0878	0.0795	0.1189
SanDiego	$\mathrm{RMSE_Y}$	0.1051	0.1375	0.0991	0.1133	0.1477	0.1060	0.1220	**0.0747**
	$\mathrm{SAD_Y}$	0.1357	0.1961	0.1219	0.1327	0.1718	0.1264	0.1664	**0.0875**
Pavia	$\mathrm{RMSE_Y}$	0.2004	0.1906	0.1607	**0.1444**	0.3492	0.1608	0.1501	0.3115
	$\mathrm{SAD_Y}$	0.3628	0.3783	0.2802	0.2718	0.8688	0.2811	**0.2652**	0.6250
HYDICE	$\mathrm{RMSE_A}$	0.2358	0.1640	0.1796	0.2424	0.2356	**0.1360**	0.1614	0.1572
	$\mathrm{SAD_M}$	0.4836	0.4707	0.1757	0.4060	0.3964	0.3788	0.1517	**0.1187**

From the perspective of unmixing, the reason for our result is that we added the loss of clustering to the loss function of unmixing, which can be regarded as a regularization term to guide similar pixels to learn similar structures in the abundance domain. In addition, we slightly increase the weight of anomalies RE, so that the abnormal pixels get more attention during the training process. Therefore, the abundance values of abnormal pixels are close to each other and close to 1.

For the estimation of endmembers, the results of the proposed method are similar to most mainstream unmixing algorithms. This shows that while focusing on anomalous targets, our proposed algorithm can also take into account the unmixing performance of other scenes. However, the unmixing performance on the Pavia dataset is not as good as other datasets. The same happens with other algorithms. The estimation of endmember EM3 has a large difference among these algorithms. It may be caused by severe endmember variability of EM3, which also makes it difficult to reconstruct the HSI by relying on a single endmember instance, resulting in generally high REs.

F. Detection Performance

The detection maps are displayed in Fig. 17. Basically, the proposed hyperspectral anomaly detector can better suppress the background detects targets with a lower false alarm rate, and can better adapt to many different scenarios.

The conventional RX algorithm is based on the assumption of the Gaussian distributed background. In most simple scenarios, RX can detect the shape of anomalies effectively as shown in the Coast, SanDiego, and Pavia datasets. However, the performance of RX will degrade when it meets complex scenarios or weak anomalies. In the HYDICE dataset, the weak anomalies are mixed with complex background, making them difficult to detect by RX.

CKRX algorithm, which is similar to the KRX, transforms each spectral pixel into a high-dimensional feature space by a nonlinear mapping function to exploit the high-order correlations between the spectral bands. We can see that for the Coast, SanDiego, and Pavia datasets, CKRX can recognize the anomalies but has limited ability to suppress the background. For the HYDICE dataset, CKRX fails to detect these anomalies, as shown in Fig. 17. These may be mainly because CKRX is sensitive to the way of clustering. In our experiments, the detection results of CKRX are unstable as the number of clusters changes. When this number is relatively set largely, some background pixels are far away from other centers of clusters, making them have high anomaly values. However, fewer clusters will degrade the ability of detection.

LSMAD, LSMD-MoG, ADLR, SaFra, AHMID, and DSR-ADNR are all representation-based HAD algorithms. Specifically, these methods represent the HSI as the combination of background and anomaly. Their common idea is to separate the sparse anomaly from the background.

LSMAD and LSMD-MoG are based on the low-rank representation of the background. However, when the background cannot be fully represented as the low-rank part, background, and anomaly will be difficult to separate. Hence, we can see that the incompletely separated background components exist in the detection maps of LSMAD and LSMD-MoG for the SanDiego and HYDICE datasets, which leads to a higher false alarm rate.

ADLR exploits the MVCNMF to unmix the spectra obtain distinctive abundance vectors, and further decompose the abundance as the combination of another low-rank representation and sparse anomaly part. However, the detection performance of ADLR is mainly depended on the result of MVCNMF. Under the premise that the HU result is not ideal, the error generated by the HU process will be accumulated in the HAD process. Especially for the Pavia dataset, we can observe from the abundance map that the EM1 (bridge) and EM2 (vehicles) are not separated. The abundance values of these two materials are very close. In the detection map, the pixels of the bridge are highly mixed with vehicles, which leads to a high false alarm rate.

SaFra and DSR-ADNR both represent the background as the combination of background atoms and construct the background dictionary. SaFra constructs the dictionary by fusing pure pixel index (PPI) and RX [34], while DSR-ADNR adopts k-means and detects low RX scores pixels

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

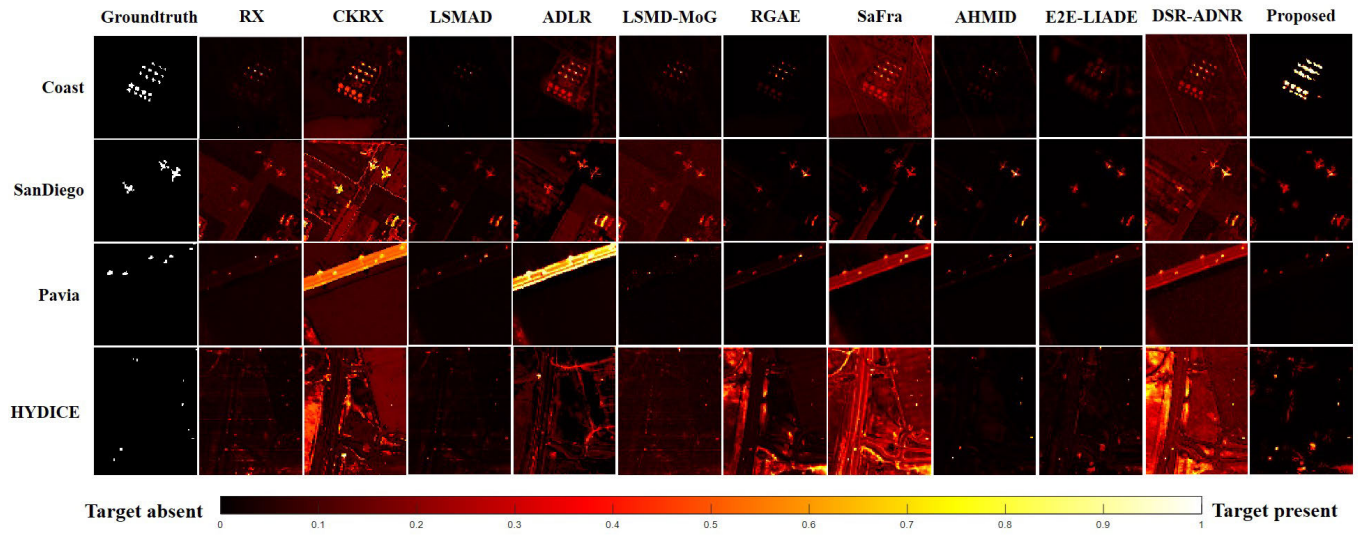


Fig. 17. HAD maps on four datasets.

TABLE III
DETECTION PERFORMANCE FOR THE ALGORITHMS ON THE EXPERIMENTAL DATASETS

Dataset	Metrics	RX	CKRX	LSMAD	ADLR	LSMD-MoG	RGAE	SaFra	AHMID	E2E-LIADE	DSR-ADNR	Proposed
Coast	AUC$(P_d, P_f) \uparrow$	0.9951	0.9973	0.9904	0.9991	0.9813	**0.9992**	0.9979	0.9545	0.9765	0.9948	0.9951
	AUC$(P_f, \tau) \downarrow$	0.0133	0.0195	0.0031	0.0309	0.0153	**0.0027**	0.1109	0.0095	0.0087	0.0783	0.0239
	AUC$(P - R) \uparrow$	0.7613	0.7807	0.7147	0.9582	0.6385	**0.9628**	0.9000	0.3966	0.2850	0.7953	0.7253
SanDiego	AUC$(P_d, P_f) \uparrow$	0.9403	0.9819	0.9659	0.7755	0.9239	**0.9921**	0.9482	0.9841	0.9846	0.9739	0.9908
	AUC$(P_f, \tau) \downarrow$	0.0589	0.1240	0.0235	0.0626	0.0793	0.0075	0.0142	0.0081	**0.0042**	0.0903	0.0084
	AUC$(P - R) \uparrow$	0.1984	0.5520	0.2210	0.1807	0.1823	**0.5774**	0.3172	0.4594	0.4641	0.5461	0.5751
Pavia	AUC$(P_d, P_f) \uparrow$	0.9887	0.9922	0.9810	0.9897	0.9501	0.9914	0.9918	0.9598	0.9869	0.9914	**0.9956**
	AUC$(P_f, \tau) \downarrow$	0.0259	0.1438	0.0137	0.3040	0.0088	0.0047	0.0378	0.0076	0.0110	0.0753	**0.0003**
	AUC$(P - R) \uparrow$	0.6715	0.6936	0.6383	0.7715	0.3187	**0.7925**	0.7891	0.6336	0.6379	0.7850	0.7751
HYDICE	AUC$(P_d, P_f) \uparrow$	0.9763	0.8575	0.9901	0.9670	0.9502	0.8412	0.9702	**0.9985**	0.9807	0.9281	0.9912
	AUC$(P_f, \tau) \downarrow$	0.0380	0.1338	0.0212	0.0583	0.0510	0.1018	0.2037	**0.0105**	0.0284	0.2229	0.0142
	AUC$(P - R) \uparrow$	0.1187	0.0131	0.3676	0.0854	0.2885	0.0075	0.3680	**0.9092**	0.2540	0.1911	0.3778
Average	AUC$(P_d, P_f) \uparrow$	0.9751	0.9572	0.9818	0.9328	0.9514	0.9560	0.9770	0.9742	0.9822	0.9721	**0.9932**
	AUC$(P_f, \tau) \downarrow$	0.0340	0.1053	0.0154	0.1140	0.0386	0.0292	0.0917	**0.0089**	0.0131	0.1167	0.0117
	AUC$(P - R) \uparrow$	0.4375	0.5099	0.4854	0.4990	0.3563	0.5851	0.5936	0.5997	0.4103	0.5794	**0.6133**

as background atoms, both of which detect the pixels from the original HSI directly. However, these dictionary construction methods rely on the high confidence pure endmembers or background pixels existing in the HSI, otherwise the background will not be represented well. Also, these algorithms place no physical constraints (such as ANC and ASC) on the coefficients of the atoms, resulting in the background pixels overlapping with anomalies, which is also reflected in the detection maps.

AHMID performs well in all datasets since there is a dictionary-learning process in the algorithm. The background and anomaly are separated clearly. However, the pixel intensities of anomalies are relatively weak, especially in the Coast dataset. The edge of planes in the SanDiego dataset is also blurred. Also, AHMID detectors do not consider the spatial information of the image, which leads to their limited detection performance.

RGAE is a deep learning-based algorithm. This algorithm takes into account the local spatial information and uses a robust AE framework to extract intrinsic features of HSI. We can see the RGAE detector performs well in the first three datasets but not in the HYDICE dataset. This is due to the fact that there are other pixels except for anomalies that are difficult to reconstruct by AE in this complex scene. Most anomalies are mixed with the background visually. Hence, the RE criterion to identify the anomalies is limited in the HYDICE dataset.

As a density estimation-based HAD algorithm, E2E-LIADE adopts a traditional AE as the backbone network for feature extraction. Without any constraints, the features extracted by traditional AE lack intuitive physical meaning, which makes it lack stability and reliability to directly use the extracted features as the criteria for anomaly discrimination.

As a comparison, our proposed method uses convolutional AE as the backbone network and imposes physical constraints on the feature extraction of the network through the upstream task HU. We can observe the characteristics of anomalies from both the detection maps and the abundance maps,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

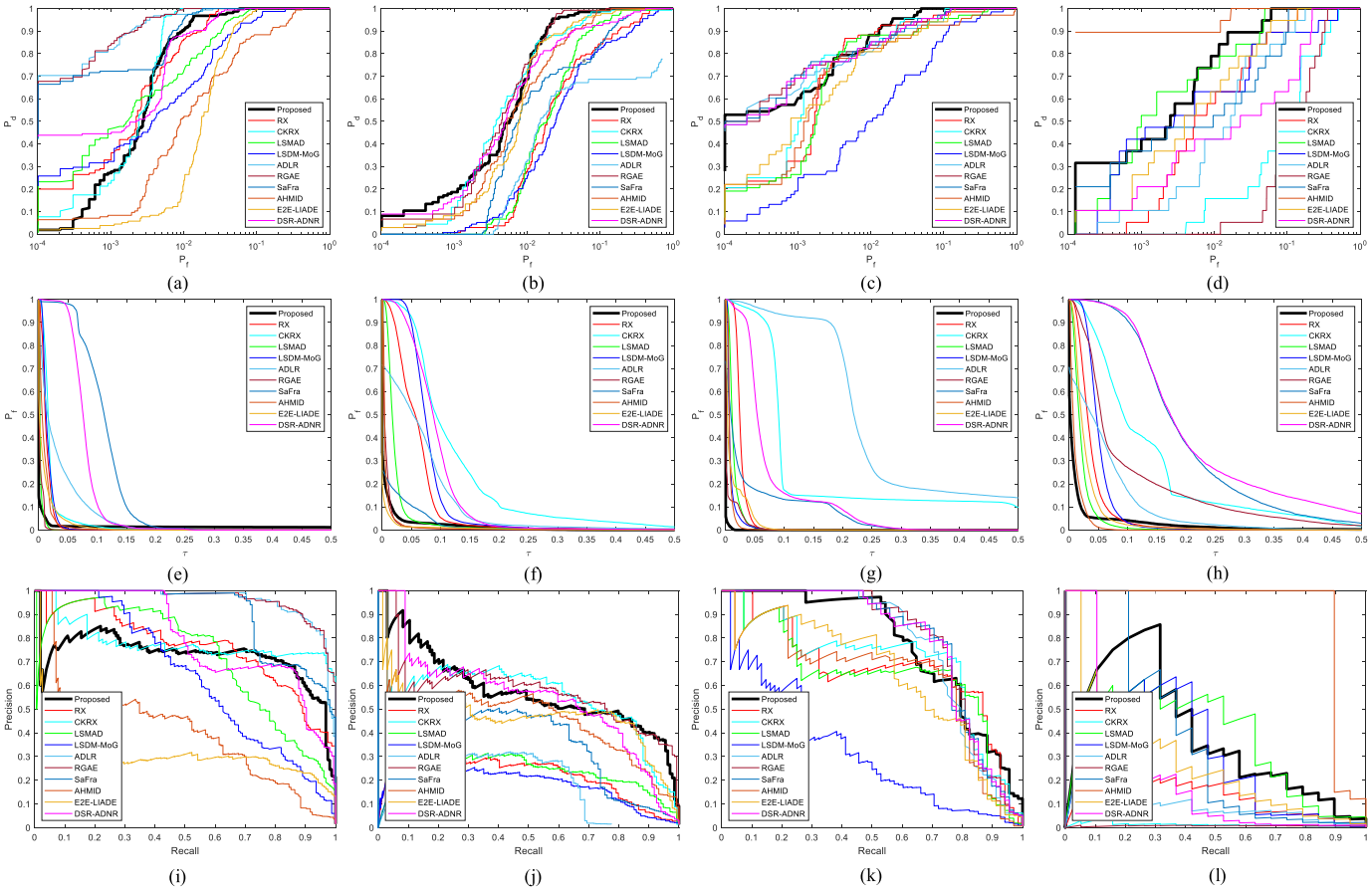


Fig. 18. First row is the ROC curves of (P_d, P_f): (a) Coast; (b) SanDiego; (c) Pavia; and (d) HYDICE. The second row is the ROC curves of (P_f, τ): (e) Coast; (f) SanDiego; (g) Pavia; and (h) HYDICE. The third row is the PR curves of (i) Coast; (j) SanDiego; (k) Pavia; and (l) HYDICE.

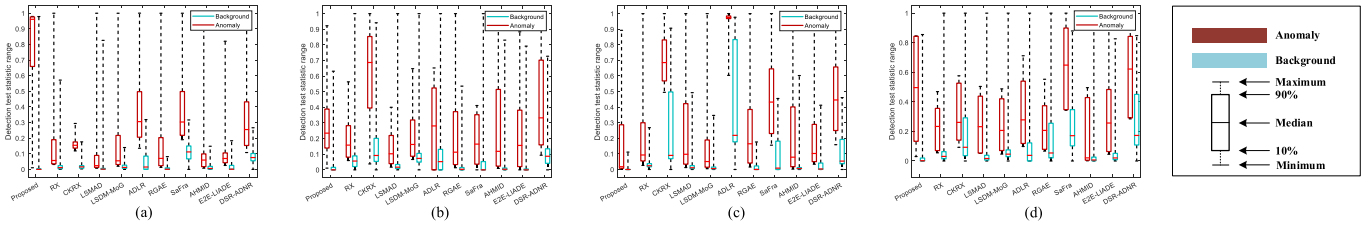


Fig. 19. Box-Whisker plot on four datasets: (a) Coast; (b) SanDiego; (c) Pavia; and (d) HYDICE.

and there is a strong connection between them. The pixel intensities of anomalies are consistent with abundance maps. Moreover, convolutional AE has a stronger local information extraction ability than traditional AE, which can also improve the detection performance.

To further evaluate the ability of the aforementioned algorithms to detect the anomalies from the background, the ROC curves and PR curves of different algorithms on all datasets are plotted in Fig. 18, and Box-Whisker plots are plotted in Fig. 19. The quantitative metrics are listed in Table III.

The ROC curves of (P_d, P_f) and PR curves of the proposed method are in the upper left of most of the algorithms, which demonstrates the excellent detection performance of the proposed method. The proposed method also gets the best average AUC(P_d, P_f) and AUC$(P-R)$ score. The ROC curve of (P_f, τ) can reflect the ability of background suppression. We can see that the ROC curves of (P_f, τ) of the proposed method are in the lower left of most algorithms and get the second best average AUC(P_f, τ) score.

The box-plot shown in Fig. 19 reveals the separability of anomalies from the background pixels. The maximum, minimum, and median of detection values are calculated, excluding the biggest 10% and the smallest 10%. The position of the boxes reflects the tendency and compactness of the distribution of the pixels. A good detection performance means that the gap between background and anomaly boxes is relatively big. Also, the smaller median and variance of background means a more powerful ability to suppress the background.

For the Coast dataset shown in Fig. 19(a), the background and the anomaly of the proposed method have the greatest separability, and the median of anomalies is close to 1. Also, the variance of background is smaller than other methods. For RX, LSMAD, LSDM-MoG, RGAE, AHMID, and E2E-LIADE, the separability of the anomaly box and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE RESULTS FOR ABLATION STUDY

Dataset	Metrics	Baseline	A	B	C	BC	AB	AC	ABC
Coast	AUC$(P_d, P_f) \uparrow$	0.9759	0.9940	0.9829	0.9780	0.9663	0.9601	**0.9954**	0.9951
	AUC$(P_f, \tau) \downarrow$	0.0431	0.0221	0.0167	0.0060	0.0094	**0.0002**	0.0116	0.0239
	AUC$(P - R) \uparrow$	0.3126	0.6198	0.4805	0.3411	0.5580	0.5981	**0.7415**	0.7253
SanDiego	AUC$(P_d, P_f) \uparrow$	0.9511	0.9670	0.9867	0.9506	0.9872	0.9814	0.9381	**0.9908**
	AUC$(P_f, \tau) \downarrow$	0.0313	0.0178	0.0124	0.0078	**0.0042**	0.0048	0.0052	0.0084
	AUC$(P - R) \uparrow$	0.1383	0.1730	0.5172	0.3376	0.4975	0.4288	0.1173	**0.5751**
Pavia	AUC$(P_d, P_f) \uparrow$	0.9433	0.9383	0.9556	0.7741	0.9812	**0.9959**	0.9552	0.9956
	AUC$(P_f, \tau) \downarrow$	0.0449	0.0030	**0.0002**	0.0008	0.0118	0.0003	**0.0002**	0.0003
	AUC$(P - R) \uparrow$	0.3299	0.1930	0.3288	0.2623	0.2987	**0.7965**	0.3874	0.7751
HYDICE	AUC$(P_d, P_f) \uparrow$	0.4514	0.9892	0.8010	0.8552	0.8547	0.9910	**0.9959**	0.9912
	AUC$(P_f, \tau) \downarrow$	0.0176	0.0297	0.0161	0.0143	0.0312	0.0164	0.0201	**0.0142**
	AUC$(P - R) \uparrow$	0.0012	0.2266	0.0141	0.0193	0.0670	0.2650	0.3211	**0.3778**
Average	AUC$(P_d, P_f) \uparrow$	0.8304	0.9721	0.9316	0.8895	0.9474	0.9821	0.9712	**0.9932**
	AUC$(P_f, \tau) \downarrow$	0.0342	0.0182	0.0114	0.0072	0.0142	**0.0054**	0.0093	0.0117
	AUC$(P - R) \uparrow$	0.1955	0.3031	0.3352	0.2401	0.3553	0.5221	0.3918	**0.6133**

TABLE V
TIME CONSUMPTION(S) OF DIFFERENT ALGORITHMS ON THE EXPERIMENTAL DATASETS

Dataset	RX	CKRX	LSMAD	ADLR	LSMD-MoG	RGAE	SaFra	AHMID	E2E-LIADE	DSR-ADNR	Proposed
Coast	0.11	0.44	9.11	4.86	16.73	85.70	100.26	14.31	93.03	8.53	29.46
SanDiego	0.08	0.33	8.35	5.16	5.32	80.79	100.48	14.59	91.28	8.92	29.55
Pavia	0.04	0.21	4.52	3.45	4.38	31.62	94.20	12.01	82.26	8.53	25.57
HYDICE	0.06	0.41	5.45	8.05	4.07	35.80	79.35	23.63	76.51	6.91	28.08

the background box is not obvious. For ADLR and SaFra, the variance of background is relatively higher. In terms of the SanDiego dataset shown in Fig. 19(b), the anomaly box is overlapped with the background box obviously for RX, LSDM-MoG, and ADLR, and the gaps between the boxes of other comparison methods are also very small. For the Pavia dataset shown in Fig. 19(c), the gap between the boxes of the proposed method is not obvious. However, it still shows good performance in background suppression. For CKRX, ADLR, SaFra, and DSR-ADNR, the variance of background is higher, which is also reflected in the visual detection map in Fig. 17. For the HYDICE dataset shown in Fig. 19(d), the proposed method also shows the greatest ability to separate the background and anomalies. For CKRX, ADLR, RGAE, SaFra, and DSR-ADNR, their background variance is relatively higher. For most of the other algorithms, their anomaly boxes are overlapped with the background boxes.

G. Ablation Study

In order to evaluate the contributions of each proposed module on detection performance, we assess the detection quality of the proposed method after removing different modules, respectively, including the HU process, feature fusion, and reweighted RE strategy. For convenience, we denote these three modules as Configurations A–C. The configurations after removing different modules are described as follows.

1) *Baseline:* The baseline is the proposed UADNet without the HU process, feature fusion, and reweighted RE strategy, which means that the AE part of the whole network has no constraints of HU, also, the decoder is initialized randomly instead of by VCA-extracted endmembers. The output features of the bottleneck are clustered directly without fusing them with the PCs of HSI. The RE loss is calculated by (14) instead of (16).
2) *Configuration A:* This configuration adds the HU constraints on the AE part. Hence, the outputs of the bottleneck layer in the AE part have real physical meanings.
3) *Configuration B:* This configuration adopts PCA to get the PCs of the HSI. Then, the PCs are connected to the output feature of the bottleneck layer.
4) *Configuration C:* This configuration adopts the reweighted RE loss, formulated by (16).

Table IV summarizes the results of ablation studies. In general, the overall performance of the method with all configurations is the best. We can observe that Configuration A has the most obvious configuration to the baseline method. Both AUC(P_d, P_f) and AUC$(P - R)$ scores have improved, since Configuration A extracts physically meaningful abundance features for the detection process. Configuration B also utilizes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

the information from the original HSI to enhance the ability of detection. With the information of the abundance of PCs, Configuration C can also help to improve the detection result. However, since the reweighted hyperparameter λ_{RE} is only slightly larger than the original weight (P_{ap}/P), the improvement is not obvious for the datasets with small size anomalies like Pavia dataset and HYDICE dataset. On the contrary, the baseline with only Configuration C still has limited detection performance, since there is not any physically meaningful information from the HSI directly. For Coast and HYDICE datasets, their abundance maps have a stronger relationship with the detection map, which we can observe from their ground truth detection maps and abundance maps in Figs. 9, 15, and 17. The augmented PCs information increases the redundancy instead. Hence the performances of baseline with Configuration AC are even better than baseline with Configurations AB and ABC.

H. Running Time

In this section, we compare the running time of all the above detection algorithms on different datasets to illustrate the computational complexity. The detailed results are shown in Table V. RX simply calculates the Mahalanobis distance between the testing pixel and the mean of all pixels and performs the most efficiently. CKRX adopts the clustering strategy to reduce the computational burden. LSMAD, ADLR, LSMD-MoG, and DSR-ADNR take the similar computational time. SaFra is the most time-consuming of all datasets. Since RGAE, E2E-LIADE, and our proposed UADNet are deep learning-based algorithms, they are more time-consuming than most detectors. However, our proposed method uses a convolutional AE as the backbone of abundance feature extraction, which has fewer parameters in forward propagation and backpropagation and thus takes less time than RGAE and E2E-LIADE.

V. Discussion and Conclusion

In this article, a novel joint unmixing and anomaly detection algorithm named UADNet is proposed. We explore the relationship between unmixing and anomaly detection of these two unsupervised tasks and design a network that can process HU and HAD in an end-to-end manner simultaneously. The unmixing network projects the input HSI to the abundance domain. In addition, we use the reweighted RE as the loss of the unmixing net to enhance the role of anomalies in the HU. The abundance can provide physically meaningful features for distinguishing anomalies from the background pixels. In order to avoid the limited representation ability of low dimensional abundance, the PCs of HSI are fused with the abundance to enhance the representation ability. Then, the augmented features obtained by abundances and PCs are clustered by the clustering network. The anomalous degree of each pixel can be calculated by the GMM pdf.

Meanwhile, there is a mutual feedback mechanism between the unmixing process and the anomaly detection process. The accuracy of unmixing will directly affect the performance of clustering and anomaly detection. The clustering result can guide the HU network to learn more about discriminative abundance. The mutual feedback mechanism can balance the performances between two tasks and make both tasks reach a relatively good result. The experiments performed on several real datasets verify the rationality and superiority of the proposed UADNet algorithm.

However, there are still some limitations to the proposed method. The proposed method is based on the assumption that there is no object with the same material as the anomalous target in the scenario. (For example, in the Pavia dataset, all objects made of metal are the cars that need to be detected.) If the object with the same material exists as the background, it will be difficult to distinguish them through their spectra. Anomaly detection via abundance maps under such conditions will suffer from a high false alarm rate since the intensities of backgrounds and targets in the abundance map are close. A possible solution is to find more spatial prior information between the anomalies and backgrounds. We plan to explore it in our future work.

References

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[3] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[4] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023.

[5] M. Zhang, W. Li, X. Zhao, H. Liu, R. Tao, and Q. Du, "Morphological transformation and spatial-logical aggregation for tree species classification using hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501212.

[6] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[7] J. Zhou, C. Kwan, B. Ayhan, and M. T. Eismann, "A novel cluster kernel RX algorithm for anomaly and change detection using hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6497–6504, Nov. 2016.

[8] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[9] N. Keshava and J. F. Mustard, "Spectral unmixing," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 44–57, Jan. 2002.

[10] Y. Qu et al., "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[11] S. Song, H. Zhou, Y. Yang, and J. Song, "Hyperspectral anomaly detection via convolutional neural network and low rank with density-based clustering," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3637–3649, Sep. 2019.

[12] M. Song, Y. Zhong, A. Ma, X. Xu, and L. Zhang, "A joint spectral unmixing and subpixel mapping framework based on multiobjective optimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519217.

[13] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.

[14] K. Jiang et al., "E2E-LIADE: End-to-end local invariant autoencoding density estimation model for anomaly target detection in hyperspectral image," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 11385–11396, Nov. 2022.
[15] J. M. Bioucas-Dias and A. Plaza, "An overview on hyperspectral unmixing: Geometrical, statistical, and sparse regression based approaches," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2011, pp. 1135–1138.
[16] N. Dobigeon, J.-Y. Tourneret, C. Richard, J. C. M. Bermudez, S. McLaughlin, and A. O. Hero, "Nonlinear unmixing of hyperspectral images: Models and algorithms," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 82–94, Jan. 2014.
[17] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–275, Jul. 1999.
[18] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.
[19] N. Dobigeon, S. Moussaoui, M. Coulon, J.-Y. Tourneret, and A. O. Hero, "Joint Bayesian endmember extraction and linear unmixing for hyperspectral imagery," *IEEE Trans. Signal Process.*, vol. 57, no. 11, pp. 4355–4368, Nov. 2009.
[20] Y. Zhou, A. Rangarajan, and P. D. Gader, "A Gaussian mixture model representation of endmember variability in hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2242–2256, May 2018.
[21] J. M. Bioucas-Dias and M. A. T. Figueiredo, "Alternating direction algorithms for constrained sparse regression: Application to hyperspectral unmixing," in *Proc. 2nd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, 2010, pp. 1–4.
[22] Y. Ma, C. Li, X. Mei, C. Liu, and J. Ma, "Robust sparse hyperspectral unmixing with $\ell_{2,1}$ norm," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1227–1239, Jun. 2017.
[23] X.-R. Feng, H.-C. Li, R. Wang, Q. Du, X. Jia, and A. Plaza, "Hyperspectral unmixing based on nonnegative matrix factorization: A comprehensive review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4414–4436, 2022.
[24] B. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "Blind hyperspectral unmixing using autoencoders: A critical comparison," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1340–1372, 2022.
[25] B. Palsson, J. Sigurdsson, J. R. Sveinsson, and M. O. Ulfarsson, "Hyperspectral unmixing using a neural network autoencoder," *IEEE Access*, vol. 6, pp. 25646–25656, 2018.
[26] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, "Convolutional autoencoder for spectral–spatial hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 535–549, Jan. 2020.
[27] Q. Jin, Y. Ma, F. Fan, J. Huang, X. Mei, and J. Ma, "Adversarial autoencoder network for hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 4555–4569, Aug. 2021.
[28] Y. Yu, Y. Ma, X. Mei, F. Fan, J. Huang, and H. Li, "Multi-stage convolutional autoencoder network for hyperspectral unmixing," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 113, Sep. 2022, Art. no. 102981.
[29] S. Matteoli, T. Veracini, M. Diani, and G. Corsini, "A locally adaptive background density estimator: An evolution for RX-based anomaly detectors," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 323–327, Jan. 2014.
[30] A. Schaum, "Joint subspace detection of hyperspectral targets," in *Proc. IEEE Aerosp. Conf.*, Jul. 2004, p. 1824.
[31] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.
[32] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.
[33] S. Li, W. Wang, H. Qi, B. Ayhan, C. Kwan, and S. Vance, "Low-rank tensor decomposition based anomaly detection for hyperspectral imagery," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Quebec City, QC, Canada, Sep. 2015, pp. 4525–4529.
[34] X. Shen, H. Liu, J. Nie, and X. Zhou, "Matrix factorization with framelet and saliency priors for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504413.
[35] T. Guo, L. He, F. Luo, X. Gong, Y. Li, and L. Zhang, "Anomaly detection of hyperspectral image with hierarchical antinoise mutual-incoherence-induced low-rank representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510213.
[36] M. Wang, D. Hong, B. Zhang, L. Ren, J. Yao, and J. Chanussot, "Learning double subspace representation for joint hyperspectral anomaly detection and noise removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507517.
[37] K. Jiang, W. Xie, J. Lei, T. Jiang, and Y. Li, "LREN: Low-rank embedded network for sample-free hyperspectral anomaly detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 5, pp. 4139–4146.
[38] M. J. Carlotto, "A cluster-based approach for detecting man-made objects and changes in imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 374–387, Feb. 2005.
[39] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.
[40] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst. Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.
[41] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, Mar. 2015, Art. no. e0118432.
[42] K. A. Islam, M. Shahabuddin, C. Kwan, and J. Li, "Flood detection using multi-modal and multi-temporal images: A comparative study," *Remote. Sens.*, vol. 12, p. 2455, Jul. 2020.
[43] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6287–6304, Nov. 2017.
[44] B. Rasti, B. Koirala, P. Scheunders, and J. Chanussot, "MiSiCNet: Minimum simplex convolutional network for deep hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522815.
[45] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503914.
[46] S. Shi, M. Zhao, L. Zhang, Y. Altmann, and J. Chen, "Probabilistic generative model for hyperspectral unmixing accounting for endmember variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516915.
[47] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511314.



Wendi Liu received the B.S. degree from the Electronic Information School, Wuhan University, Wuhan, China, in 2022, where he is currently pursuing the master's degree with the Multi-Spectral Vision Processing Laboratory.

His research interests include hyperspectral imagery, computer vision, and machine learning.



Yong Ma received the bachelor's degree from the Department of Automatic Control, Beijing Institute of Technology, Beijing, China, in 1997, and the Ph.D. degree from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 2003.

From 2004 to 2006, he was a Lecturer at the University of the West of England, Bristol, U.K. From 2006 to 2014, he was with the Wuhan National Laboratory for Optoelectronics, HUST, where he was a Professor of electronics. He is now a Professor with the Electronic Information School, Wuhan University, Wuhan. His general field of research is in signals and systems. His research projects include remote sensing of the lidar and infrared, as well as infrared image processing, pattern recognition, and interface circuits to sensors and actuators.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.



Xiaozhu Wang received the B.S. degree from the School of Electronic Information, Changchun University of Science and Technology, Changchun, China, in 2013, the M.S. degree in electronic information from the Northwest University of Technology, Xi'an, China, in 2016, and the Ph.D. degree in armament Science and technology from the Northwest University of Technology, Xi'an, in 2020.

He is currently with the School of Data Science, The Chinese University of Hong Kong at Shenzhen, Shenzhen, China. His research interests include image processing and optoelectronic information processing.



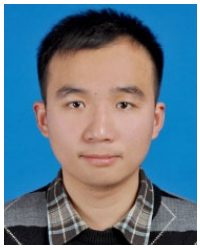
Jun Huang received the B.S. and Ph.D. degrees from the Department of Electronic and Information Engineering, Huazhong University of Science and Technology, Wuhan, China, in 2008 and 2014, respectively.

He is currently working as an Associate Professor at the Electronic Information School, Wuhan University, Wuhan. His research interests include infrared image processing and infrared spectrum processing.



Qihai Chen received the Ph.D. degree from the School of Astronautics, Harbin Institute of Technology, Harbin, China, in 2009.

He is currently the Director of the Space Institute, Tsinghua Innovation Center in Zhuhai, Zhuhai, China. His research interests include satellite image processing and navigation and control.



Hao Li received the bachelor's degree in communication engineering and the Ph.D. degree in electronic science and technology from the Huazhong University of Science and Technology, Wuhan, China, in 2006 and 2014, respectively.

From 2014 to 2015, he worked first as a Research Associate and then as a Post-Doctoral Fellow with the Center for Transport, Trade and Financial Studies, City University of Hong Kong, Hong Kong. In 2016, he joined the College of Mathematics and Computer Science, Wuhan Polytechnic University, Wuhan, where he is currently a Lecturer. His research interests include hyperspectral and infrared signal processing, as well as their applications in remote sensing, food safety, and agricultural production.



Xiaoguang Mei (Member, IEEE) received the B.S. degree in communication engineering from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 2007, the M.S. degree in communications and information systems from Huazhong Normal University, Wuhan, in 2011, and the Ph.D. degree in circuits and systems from HUST in 2016.

From 2010 to 2012, he was a Software Engineer with the 722 Research Institute, China Shipbuilding Industry Corporation, Wuhan. From 2016 to 2019, he was a Post-Doctoral Fellow with the Electronic Information School, Wuhan University, Wuhan, where he is currently an Assistant Professor. His research interests include hyperspectral imagery, machine learning, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:25:01 UTC from IEEE Xplore. Restrictions apply.